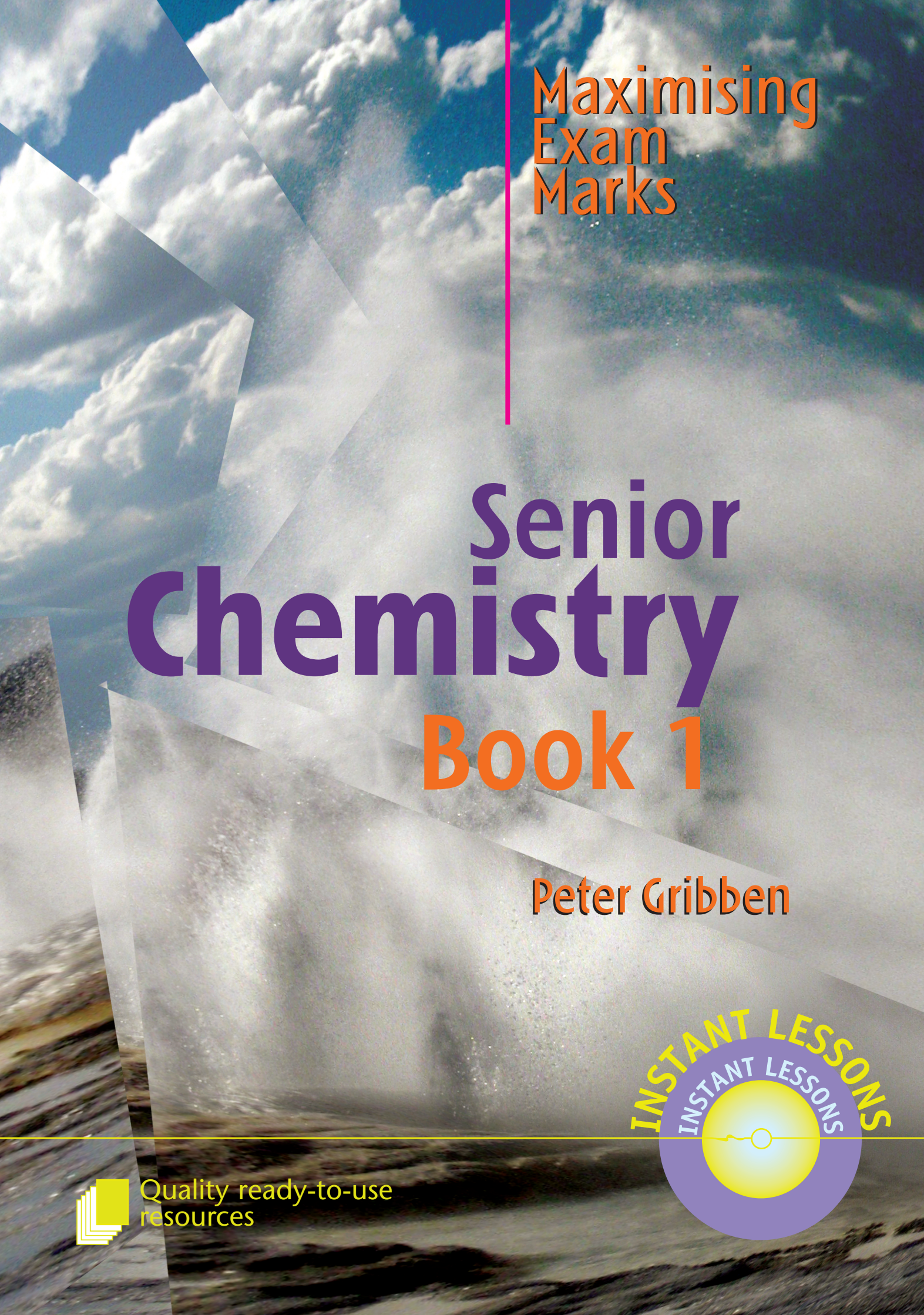
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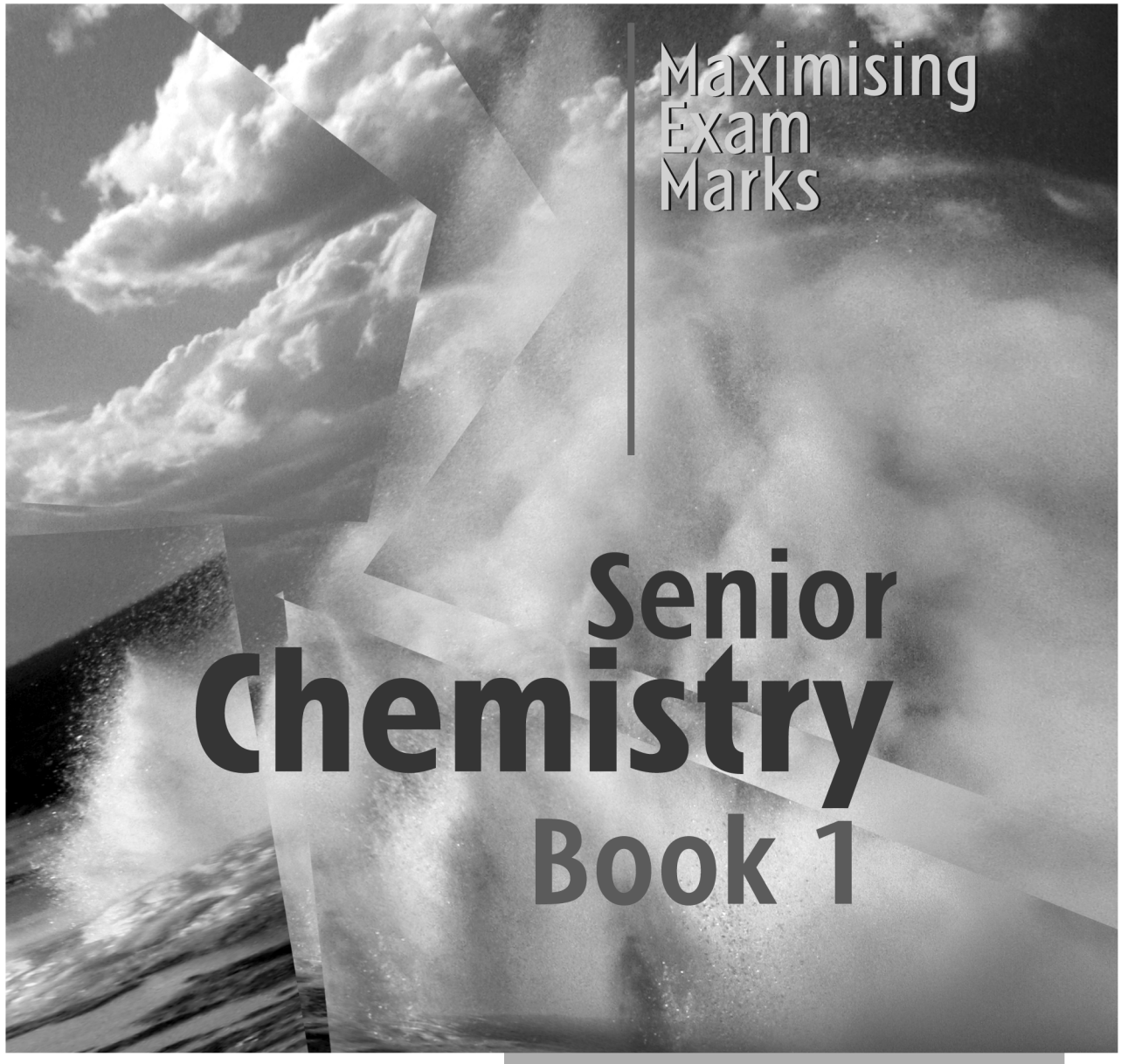

Peter Gribben

This book was commissioned by Jennifer Gough, project management was by Piper Print Media and Gail McManus Graphics designed the cover.

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ABN 50 074 266 023
108 Main Road
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ISBN 978-1-92561-014-7

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INTRODUCTION

Every year, examiners, teachers and assessors in every State produce detailed reports on how students have coped with the various questions on their exams. And every year the same errors are made by students. This series is designed to assist senior students in areas that they have traditionally found diffficult.

Questions focusing on the common errors in the context of the topic area in which they typically occur will help students avoid wasting valuable marks. Key labels mark important content, concepts and techniques:

THINGS TO KNOW – Content that students should learn and be able to apply

HINTS – Helping students approach questions correctly

TRICKS – Giving short-cuts or easier ways to attack questions

SLIPS – Providing opportunities to practise working carefully to avoid common errors

TRAPS – Providing the types of contexts in which students make common errors

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1 THE ATMOSPHERE

The atmosphere makes up only about 0.0001% of the total mass of the Earth – but without it, life as we know it would not exist.

1 Table 1 shows some details of the gases that make up the atmosphere. Copy and complete the table.

TRAP Watch out for diatomic gases.

Table 1

Gas	Formula	Percentage
Nitrogen		78.08
	O_2	20.95
Argon		0.93
	CO_2	0.036
Neon		0.0018

The atmosphere is the whole mass of air surrounding the Earth. Nitrogen makes up the around 80% of the atmosphere.

Nitrogen is an unreactive gas at room temperature and atmospheric pressure. Nitrogen can be made to react in industrial processes by using high temperatures, high pressures and special catalysts. Nitrogen does not support combustion.

Nitrogen compounds are essential to plant and animal life. However, because nitrogen is so stable, most living things cannot use nitrogen directly from the atmosphere. The only organisms that can take in elemental nitrogen are nitrogen-fixing bacteria which are found in soil, usually in or near the roots of legumes.

The nitrogen cycle (Figure 1) shows how nitrogen and nitrogen compounds move between the biosphere and various organisms.

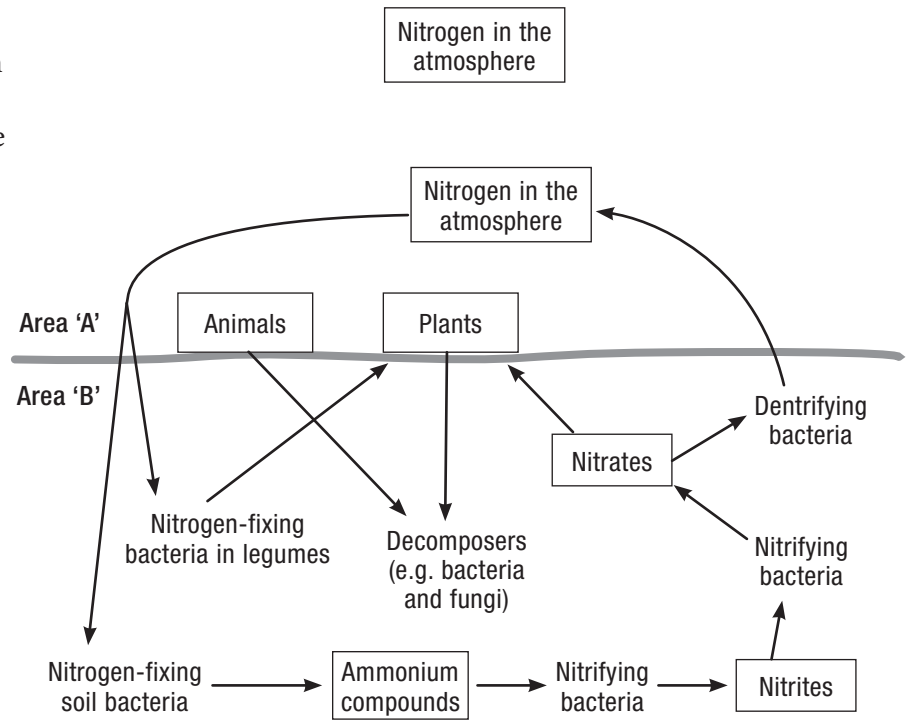

Figure 1

2 What is the biosphere?

HINT 'Bio' refers to living things.

3 Refer to figure 1.

a What is area A?

b What is area B?

4 Table 2 shows the names of some nitrogenous species and the formulas of these species, but not in the correct order. Draw up a table listing each species with the correct formula beside it.

Table 2

Species name	Formula
Ammonium	NO_2^-
Nitrate	NH_4^+
Nitrite	NO_3^-

THINGS TO KNOW Make sure you know the difference between a compound and a species.

5 Identify the roles in the nitrogen cycle of the following organisms.

a nitrifying bacteria

b nitrogen-fixing soil bacteria

c decomposers

d denitrifying bacteri

6 Figure 1 doesn't show any direct interaction between animals and plants. Describe the possible links between these groups.

7 Write a definition for *atmosphere*.

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2 OXYGEN

THINGS TO KNOW Oxygen (O_2) is a very reactive gas and supports combustion.

Oxygen was first prepared in 1774 by Joseph Priestly. Priestly used apparatus similar to that shown in figure 1. He heated solid mercury (II) oxide which decomposed into liquid mercury and a gas. This gas, which we now know to be oxygen, allowed things to burn in it more vigorously than they did in air. It could also be breathed by mice.

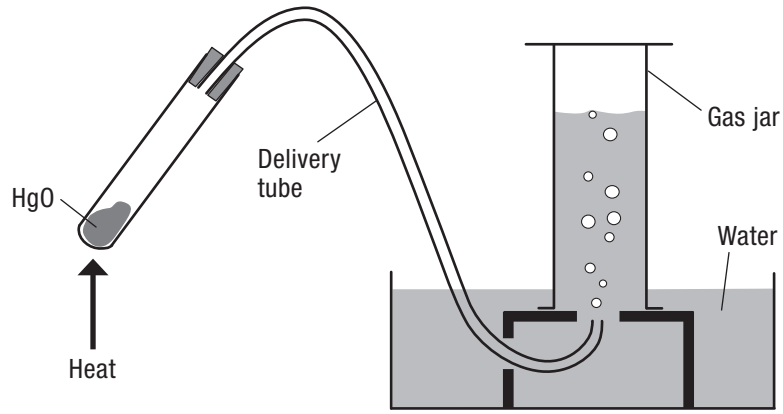

Figure 1

1 Describe how the gas is collected using the apparatus in Figure 1.

TRICK Diagrams in Chemistry often show the process proceeding from left to right.

2 Write a balanced equation for the reaction Priestly used.

SLIP Be careful in balancing the equation. Include symbols of state.

3 Give a reason for scientists no longer using this method to generate oxygen.

The most common way to generate oxygen in the school laboratory is by the decomposition of a solution of hydrogen peroxide (H_2O_2) with solid manganese dioxide (MnO_2) catalyst.

4 Write a balanced equation for the reaction described above.

5 Define *catalyst*.

Figure 2 shows one way to burn substances. A deflagrating (combustion) spoon consists of a metal rod fitted with a lid and a spoon on the end. The substance is placed on the spoon, heated and then quickly placed in a gas jar. The lid seals the contents from the outside atmosphere.

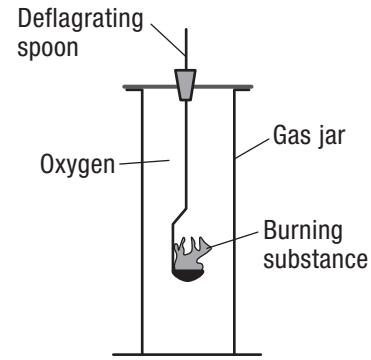

Figure 2

6 How would a substance burn in oxygen as opposed to burning in air? Explain your answer.

A teacher demonstrated burning solid sulfur using the apparatus shown in figure 2. She included some moist litmus paper at the bottom of the gas jar. The sulfur dioxide produced turned blue litmus to red.

SLIP Be careful in balancing the equation. Include symbols of state.

7 Write a balanced equation for the combustion of sulfur.

8 What information about sulfur dioxide is given by the moist litmus?

9 Oxygen is only slightly soluble in water: 0.003 93 g per 100 g of water at 25°C at 1 atm pressure.

a An aquarium tank is 60 cm × 40 cm × 40 cm. Calculate the volume of the tank.

b What is the total mass of oxygen that could be dissolved in the tank if it were full of water? Assume that 1 cm^3 (1 mL) of water has a mass of 1 g.

Most oxygen present in the atmosphere is in the form of diatomic molecules; that is, as O_2. However, it can also exist as O_3, ozone. High voltage electrical discharges can form ozone from 'normal' oxygen. It is also generated in the upper levels of the atmosphere by the action of ultraviolet light on oxygen.

10 The media frequently reports concerns about the 'ozone layer'. Outline what this layer does and why there are concerns about it.

TRAP Don't confuse the 'Greenhouse effect' with the 'ozone layer'.

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3 CARBON DIOXIDE

THINGS TO KNOW

Carbon dioxide (CO_2) is a colourless and odourless gas that doesn't support combustion.

Carbon dioxide isn't toxic in low concentrations, but when it reaches around 10% of air it can cause humans to lose consciousness. It is denser than air and is more soluble in cold water than warm water. Carbon dioxide forms an acidic solution in water and it turns bicarbonate indicator from to red to yellow. Carbon dioxide is cycled through the life processes of photosynthesis and respiration.

1 What is meant by *toxic*?

Carbon dioxide reacts with lime water (calcium hydroxide solution, $Ca(OH)_2$) to form a cloudy suspension of calcium carbonate ($CaCO_3$) and water.

2 Write a balanced equation for the reaction described above.

3 Carbon dioxide is cycled through the life processes of photosynthesis and respiration. Copy figure 1 and show the directions of photosynthesis and respiration.

$$6CO_{2(g)} + 6H_2O_{(l)} \rightleftharpoons C_6H_{12}O_{6(s)} + 6O_{2(g)}$$

Figure 1

THINGS TO KNOW

The actual processes are more complex than these equations show.

Large bodies of water play a significant role in the carbon cycle (figure 2). Seas and oceans contain huge quantities of dissolved carbon dioxide.

4 What will happen to the dissolved carbon dioxide in seas and oceans if water temperature increases? Explain your answer.

5 Study the carbon cycle in figure 2.

a Label the air, land and sea.

b Define *biomass*.

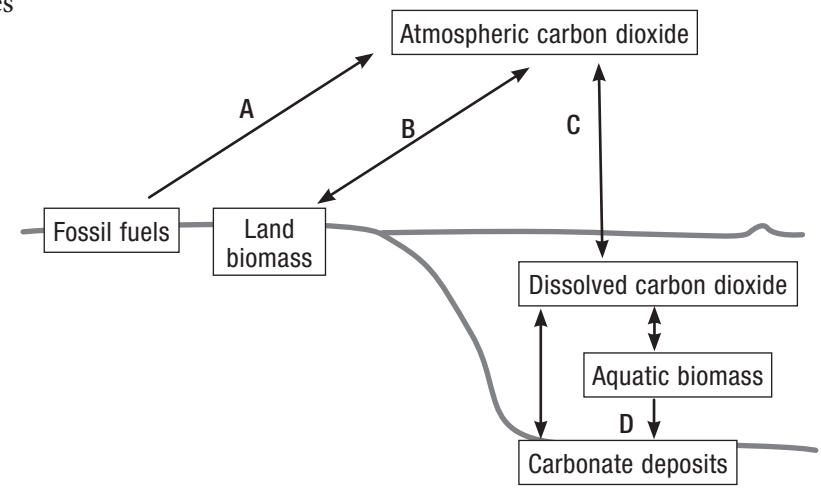

Figure 2

6 Match the letters A to D from figure 2 to the correct processes below.

- When marine organisms die, their shells and body parts sink to the ocean floor where they can accumulate as carbonate-rich deposits.
- Carbon dioxide passes between the air and the ocean.
- Petrol, coal and oil are burned to supply energy.
- Respiration and photosynthesis occur simultaneously.

7 Copy the following passage and complete it using these words: *biological, chlorophyll, light, live, metabolism, ocean, oxygen, phytoplankton, sink, unicellular.*

Marine _____ play a major part in the carbon cycle. They account for about 50% of the global _____ uptake of carbon dioxide. Phytoplankton are _____ organisms. Near the ocean surface, they use _____ energy to convert carbon dioxide into organic molecules for building cellular structures and to drive their _____. Phytoplankton contain _____ which enables them to photosynthesise. They produce much of the _____ present in the Earth's atmosphere, around 50% of all that is produced by plant life. Individual phytoplanktons only _____ for a few days. When they die, they descend to the bottom of the _____. Consequently, over a very long period of time, the ocean has become the primary storage _____ for atmospheric carbon dioxide.

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4 ATMOSPHERIC POLLUTION

In modern agriculture, the use of pesticides is necessary. Aerial spraying ('crop dusting') is used to cover large or inaccessible areas. However, not all of the chemicals sprayed reach their intended sites. It has been estimated that about 85% of pesticides delivered in this way can drift tens of kilometres from the point of delivery.

Many agricultural operations can generate particulate matter into the atmosphere during hot, dry weather. Methane (CH_4) gas is produced by many agricultural activities – the two major sources are rice production and ruminant livestock. In one day, a sheep can produce approximately 30 litres of methane and a dairy cow up to 200 litres.

THINGS TO KNOW

Pollution is the presence of unwanted substances especially those caused by humans. Air pollution can be caused by many human activities including agriculture, industry, transport and the production of energy.

People who live in rural areas are not exposed to the same pollutants as those who live in cities, but they have to contend with nuisance odours and other more toxic airborne pollution.

1 What is meant by *a nuisance odour*?

2 What is meant by *particulate matter*?

3 Assuming that there are 28.6 million sheep in Australia, calculate how many litres of methane they will produce in a year.

HINT

Show your working when carrying out calculations so you can check your answers more easily.

4 If methane has a density of 0.68 g L^{-1}, calculate the mass of methane produced by these sheep in a year.

The Greenhouse Effect is the heating of the Earth due to the presence of Greenhouse gases. Methane is a Greenhouse gas, which means it contributes to the Greenhouse Effect.

Without the Greenhouse Effect, the Earth's mean temperature would be around 30°C less than it is, and the daily temperature would fluctuate much more than it currently does. However, many people think that continued emissions of Greenhouse gases by human activity will increase the Greenhouse Effect and cause Global Warming – the increase of the Earth's mean temperature. Figure 1 shows the relative percentages of Greenhouse gases in Earth's atmosphere.

Relative percentages of Greenhouse gases in Earth's atmosphere

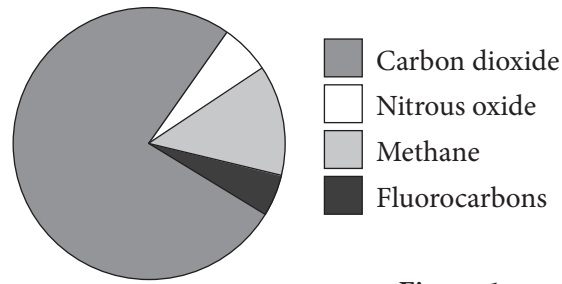

Figure 1

5 Referring to figure 1, name the Greenhouse gas present in the greatest quantity.

6 Carbon dioxide is put into the atmosphere by respiration and combustion. Name a process that removes carbon dioxide from the atmosphere.

7 Deforestation and human use of fossil fuels have been blamed for increasing levels of carbon dioxide. Name three fossil fuels.

Some Greenhouse gases have a greater effect (i.e. per mass present) than others. This is called the 'Greenhouse multiplier'. Carbon dioxide is assigned a value of 1.0, methane 21.0, nitrous oxide 310 and chlorofluorocarbons (on average) 6100.

TRAP

The Greenhouse effect should not be considered completely negative. Without it the temperature on Earth would fluctuate wildly. We should be concerned about an increase in the Greenhouse effect.

8 Nitrous oxide is produced by nitrogen-based fertilisers and internal combustion (car) engines. Calculate which of the following would contribute most to global warming. Show your working.

- 500 tonnes of carbon dioxide
- 28 tonnes of methane
- 1.67 tonnes of nitrous oxide
- 0.9 tonnes of CFCs.

HINT

The contribution will be the product of the mass times 'Greenhouse multiplier'.

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Chlorofluorocarbons (CFCs) are relatively unreactive substances that are used as non-stick coatings (e.g. PTFE, polytetrafluoroethylene, also known as TEFLON™), refrigerants and aerosol propellants. CFCs are fluorocarbons. They are being phased out because, when released, they reach the upper atmosphere where they deplete (reduce) the amount of ozone (O_3). CFCs have a dual polluting role – they are also Greenhouse gases.

9 What does an aerosol propellant do?

10 Which element makes up ozone?

11 Name three elements present in CFCs.

Ozone is formed in the upper atmosphere, 15–55 km above the surface, by the action of ultraviolet (UV) radiation on oxygen molecules. Ozone is an unstable compound and rapidly breaks down into oxygen. Ozone can be toxic to humans if concentrations are high enough. It has been used for many years to purify water and remove odours.

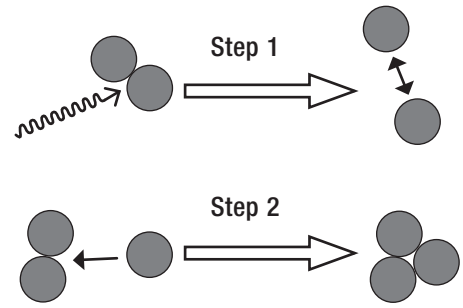

Figure 2

12 Use figure 2 and the previous information to describe how ozone is formed.

HINT Oxygen is diatomic, ozone is triatomic.

Acid rain has damaged forests and rendered lakes too acidic for aquatic life to survive. Acid rain is rain that has become acidic due to airborne pollutants. Power stations that use coal as a fuel are major contributors to atmospheric pollution. Sulfur is present in coal (and in other fossil fuels). When burned, it produces sulfur dioxide (SO_2) which is an irritant to the respiratory system and a cause of 'acid rain.' The sulfur dioxide can dissolve in water to form sulphurous acid (H_2SO_3). Petrol burned in internal combustion (car) engines can produce nitrogen dioxide which can then dissolve in water to form a mixture of nitrous (HNO_2) and nitric acids. Ozone can also be created.

13 Copy the following sentences and use these words to complete them:
hole, increase, life, screens, stratospheric.

The ozone layer _____ out a large proportion of harmful UV radiation. If it wasn't for _____ ozone, _____ as we know it wouldn't be possible on Earth. CFCs have caused a _____ in the ozone layer, which has led to an _____ in the amount of UV radiation reaching parts of Earth.

14 Write a balanced equation for the formation of sulphurous acid from sulfur dioxide and water.

TRAP Don't forget to put (aq) next to the formula when a substance is dissolved in water.

15 Give the formula for nitric acid.

16 What is meant by *aquatic life*?

Smog first appeared in some cities hundreds of years ago. Smog was originally formed by smoke from coal coalescing (joining) with fog. Today it can also include oxides of nitrogen (NOX), volatile organic solvents, carbon monoxide (CO), ozone and sulfur dioxide, with soot and ash particles. Sunlight can provide energy for these pollutants to react with each other and form even more dangerous photochemical smog, which can cause major respiratory problems.

17 What does the word *volatile* mean when referring to organic compounds?

18 What is meant by *major respiratory problems*?

19 What is the origin of the term *smog*?

20 What is the energy source that produces photochemical smog?

21 Which class of compounds is represented by NOX?

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5 GASES AND GAS LAWS

The kinetic molecular theory is very useful for explaining the properties of gases. The fundamentals on which the kinetic theory of gases is based (i.e. describing an ideal gas) are listed below.

- Gases consist of minute, discrete particles (usually molecules) in continuous, chaotic motion and with high speeds.
- The average speed of motion increases with the absolute temperature, which is thus some measure of the average energy of motion, i.e. kinetic energy. At absolute zero (0 K), movement from place to place, i.e. energy of translation, is at a minimum.
- Collisions between molecules and bombardment of the walls of the containing vessel occur, the latter is the cause of gaseous pressure, which is force per unit area.
- The collisions are assumed to be perfectly elastic without any loss of energy. Otherwise, the pressure of a given amount of gas would alter over time.
- The distances between the molecules are large in comparison with the size of the molecules. At low pressures, molecules may even be considered as point masses, i.e. have negligible volume.
- At low pressures, the forces of attraction between molecules are insignificant.

TRAP At absolute zero, particles have **not** completely stopped moving – they are just moving very, very slowly.

HINT When you come across new chemical terms, add them to your glossary.

1 Match each term with the correct meaning.

Terms: *kinetic, minute, discrete, chaotic, translation, negligible, collision, elastic*

a Coming into sudden contact
b Resumes original state
c Very small
d Insignificant
e Separate
f Movement from one place to another
g To do with movement
h Disordered

BOYLE'S LAW

Around 1662, Robert Boyle published the results of a set of experiments that investigated the relationship between the pressure and volume of a fixed mass of gas at a constant temperature. Figure 1 summarises Boyle's Law.

2 Referring to figure 1, outline the relationship between the pressure and volume of a gas when mass and temperature are kept constant.

TRICK When looking at relationships shown by a graph, ask yourself, 'What happens to Y when X changes?'

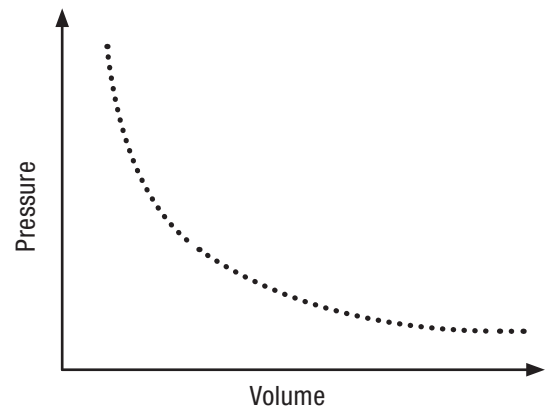

Figure 1

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Figure 2 shows one type of apparatus commonly used to demonstrate Boyle's Law. Note that this is not the actual type of apparatus used by Boyle.

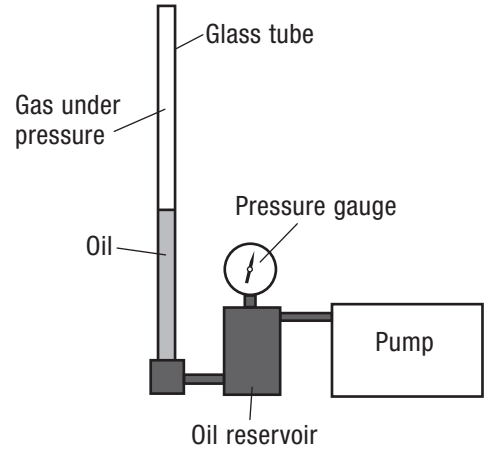

Figure 2

3 Outline how the apparatus in figure 2 would be used to illustrate Boyle's law.

Boyle's Law states that no matter how pressure (P) and volume (V) are varied, as long as mass and temperature remain constant their product, $P \times V$, will remain constant. Another way of expressing this is:

$PV = k$ (a constant)

or $P_1V_1 = P_2V_2$ where P_1 and V_1 refer to the initial conditions (before any changes) and P_2 and V_2 refer to the final conditions (after changes)

The volume of a fixed mass of gas, at constant temperature, is **inversely proportional** to its pressure.

$$P \propto \frac{1}{V}$$

$P \times V =$ a constant

TRICK When you see equations in the Gas Law, ensure that you can translate them from mathematical symbols to words and vice versa.

4 Copy the passage below and complete it using these words:
collide, decreases, fill, halved, increased, pressure, temperature, volume.

A given mass of gas will diffuse readily to _____ any container in which it is placed; the pressure of the gas _____ as the space becomes larger. Gas _____ is caused by the impact of the moving molecules on the sides of the container. If the _____ of a fixed mass of gas in a sealed container at constant _____ is reduced, the pressure will rise. This is because the molecules, which are still moving at the same speed, will _____ with the sides of the container more often. If the volume is _____, the pressure will double. If the pressure is _____ then the volume will decrease.

5 A sample of gas takes up 200 mL at 150 kPa. The pressure is changed to 275 kPa. The temperature is not changed. What would be the volume of the gas? Show your working.

THINGS TO KNOW In Chemistry, the two most common units used to measure pressure are atmospheres (atm) and kilopascals (kPa). 1 atmosphere pressure = 101.3 kPa

SLIP Ensure you use the correct equation and show units.

CHARLES' LAW

Around 1787, Jacques Charles investigated the relationship between the temperature and volume of a gas. Figure 3 summarises the results of an experiment investigating Charles' Law.

6 Name two variables that would need to be kept constant in Charles' investigation.

7 Referring to figure 3, outline the relationship between temperature and volume of a gas when mass and pressure are kept constant.

TRICK When looking at relationships shown by a graph, ask yourself, 'What happens to *Y* when *X* changes?'

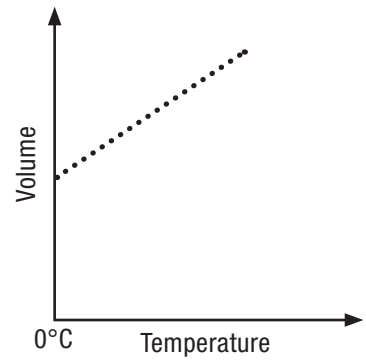

Figure 3

THINGS TO KNOW When dealing with calculations involving gases, the Kelvin scale is used. To change degrees Celsius (°C) to Kelvins (K), add 273 to the Celsius figure. Note that the actual value is 273.15.

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Figure 4 shows one type of apparatus commonly used to demonstrate Charles' Law. Note that this is not the type of apparatus actually used by Charles.

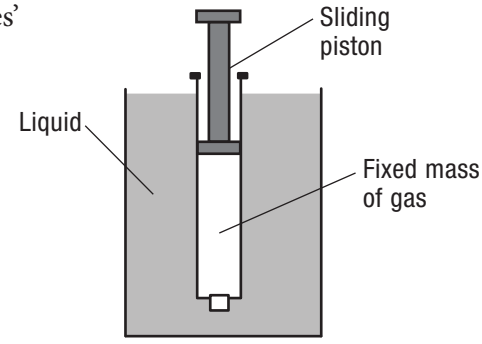

Figure 4

8 Outline how the apparatus in figure 4 could be used to illustrate Charles' Law. Also outline any limitations of the equipment.

HINT What will happen to the volume of the gas when the temperature of the liquid changes?

9 Copy and complete the passage, using these words:
collide, constant, fixed, kinetic, raised, rapidly, relationship, volume.

If we keep the _____ of a gas constant and vary the temperature, then the pressure will also vary. If the temperature is _____ the pressure will also increase. The heat energy supplied to the gas is changed to _____ energy and so the particles move more _____. The gas particles will then _____ more often with the sides of the container and thus increase the pressure. If the temperature is dropped and the volume kept _____ then the pressure will also drop. If pressure is kept constant but the volume is allowed to change then, when heated, a _____ mass of gas will expand. This _____ is set out in Charles' Law.

The volume of a fixed mass of gas at constant pressure is directly proportional to its temperature in Kelvins. We can express Charles' Law mathematically as shown in Figure 5.

$$V \propto T$$

$$\frac{V}{T} = \text{a constant}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Figure 5

10 Explain what the equation $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ means.

11 The volume of a fixed mass of gas is 18.5 L at 15°C. Calculate the volume of the gas at 30°C.

TRAP Remember to change degrees Celsius to Kelvins, i.e. add on 273 to ^{0}C to get K.

12 A sample of gas occupied 0.5 L at 41°C. What would the temperature need to be for the gas to have a volume of 0.35 L?

Figure 6 shows what happens to the volume of a sample of gas if the temperature is allowed to decrease.

13 Describe what happens to the motion of individual gas particles as the temperature decreases.

HINT Remember your basic kinetic theory.

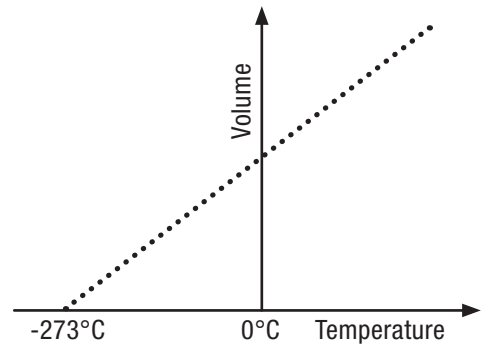

Figure 6

Because both temperature and pressure can affect the volume of gases, we need to specify under what conditions the volume of a gas is measured. One standard set of conditions chosen is 0°C (273 K) and 101.3 kPa (1 atm). At this **S**tandard **T**emperature and **P**ressure (STP), one mole of any ideal gas occupies 22.41 litres.

These conditions are also referred to as Standard Laboratory Conditions (SLC, 25°C, 101.3 kPa), where one mole of an ideal gas occupies 24.47 L.

HINT It is a good practice to ensure all temperatures are in Kelvins when carrying out calculations involving the gas laws.

14 Write down the conditions specified for STP for your Chemistry course.

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6 GAS LAWS – COMBINED AND IDEAL

We can combine Boyle's Law and Charles Law to get the equation shown in figure 1. This often referred to as the **Combined Gas Law**. Check what your textbook or syllabus calls it. This equation can be used to calculate the pressures, volumes or temperatures of gases under various conditions.

Combined Gas Law

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

Figure 1

1 Explain what is meant by the equation shown in figure 1.

WORKED EXAMPLE

A given mass of gas occupies a volume of 4 litres at 27°C.
It is heated at a constant pressure to 177°C.
What volume would it then occupy?

Answer:

P_1 and P_2 don't change ('at constant pressure') so use $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

$V_1 = 4$ L T_1 (27 +273) =330 K $V_2 = ?$ $T_2 = (177 + 273) = 450$ K

Hence $V_2 = \frac{V_1 \times T_2}{T_1} = \frac{4 \times 450}{330} =$ **5.45 L**

HINT

- In gas calculations use these units:
  - Pressure: kilopascals (kPa) or atmospheres (atm)
  - Volume: litres (L)
  - Temperature: Kelvins (K)
- Don't mix units for pressure.

SLIP Follow the sequence shown in the worked example. Check units and show working.

2 A sample of gas has a volume of 0.46 litres measured at 40°C and 199.0 kPa. What would be the volume of this gas at 0°C and 101.3 kPa?

3 A sample of gas has a volume of 5.00 litres measured at 20°C and 101.3 kPa. What would be the pressure of this gas at 50°C and a volume of 10.0 litres?

Figure 2 shows a derivation of the previous equation that includes the number of moles (n) of the gas present and the ideal gas constant (R). R has a value of 8.31 $JK^{-1}mol^{-1}$ when working in kilopascals. (If using atmospheres as a unit of pressure it is 0.082 $JK^{-1}mol^{-1}$). This equation is generally referred to as the **Ideal Gas Law**.

Ideal Gas Law

$$PV = nRT$$

Figure 2

WORKED EXAMPLE

Part a: A sample of gas occupies 12.05 litres at a pressure of 58.8 kPa and 480 K.
How many moles of this gas are present?

Answer:

$PV = nRT$

$P = 58.8$ kPa $V = 12.05$L $n = ?$
$R = 8.314$ $JK^{-1}mol^{-1}$ $T = 480$ K

Rearranging the equation $PV = nRT$ gives $n = \frac{PV}{RT}$

$n = \frac{58.8 \times 12.05}{8.314 \times 480} = \frac{708.5}{3\,990}$

= **0.178 moles**

Part b: This sample of gas had a mass of 7.47 g.
What is the molar mass of this gas?

Answer:

Molar mass = mass present ÷ number of moles

$M = m \div n$

7.47 g ÷ 0.178 = 41.2 g

Hence one mole = **41.2 g**

$$n = \frac{m}{M}$$

number of moles = mass in grams ÷ molar mass

Figure 3

4 Calculate the number of moles of a sample of gas with a volume of 1.25 L at 25°C and 1200 kPa.

5 0.46 g of a gas occupies a volume of 250 mL at 26°C and 110 kPa.
What is the molar mass of this gas?

TRAP Make sure you know how to convert masses to moles and vice versa. Check figure 3 if you need reminding.

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7 MOLAR VOLUME OF GASES

In 1811, Amadeo Avogadro proposed that equal volumes of ideal (or 'perfect') gases at the same temperature and pressure contain the same number of molecules. Avogadro did not come up with the idea of the mole or even Avogadro's number (6.02×10^{23}). He did provide the foundation for future scientists to develop these concepts.

THINGS TO KNOW

A mole of a substance contains 6.02×10^{23} particles of that substance.

1 Outline the general physical properties of gases.

From Avogadro's hypothesis, we can state that a mole of any gas should occupy the same volume under the same conditions. Usually the volume of a mole of gas under the standard conditions is quoted; for example, 1 mole of gas at 101.3 kPa (1 atm) and 0°C occupies 22.4 litres and at 101.3 kPa and 25°C occupies 24.5 litres.

2 State the value(s) of the standard conditions used in your course.

The number of moles of a gas can be calculated by dividing the volume present by the molar volume, as shown in figure 1.

$$n = \frac{v}{V}$$

number of moles = volume in litres ÷ molar volume

Figure 1

WORKED EXAMPLE

How many moles of gas are present in 49.0 L at 101.3 kPa and 0°C?

Answer:

$n = v \div V$
$= 49.0 \text{ L} \div 22.4 \text{ L}$
$= \mathbf{2.188}$

You can also calculate the volume if you know the number of moles present.

WORKED EXAMPLE

What volume would 0.33 moles of gas at 101.3 kPa and 25°C occupy?

Answer:

$v = n \times V$
$= 0.33 \times 24.5 \text{ L}$
$= \mathbf{8.085 \text{ L}}$

If you know the mass and volume of sample you can calculate **molar mass**.

WORKED EXAMPLE

8.4 g of a gas at 101.3 kPa and 25°C occupies 6.42 litres. What is the molar mass of the gas?

Answer:

First calculate the number of moles (n):
$n = v \div V$
$= 6.42 \text{ L} \div 24.5 \text{ L}$
$= 0.262$ moles
Then calculate molar mass (M):
$M = m \div n$
$= 8.4 \text{ g} \div 0.262$
$= \mathbf{32.1 \text{ g}}$

3 Calculate the number of moles of gas present at 101.3 kPa and 0°C for each of the following.

a 3.7 L

b 49.5 L

c 250 mL

TRAP

Don't forget to use the correct formula (Figure 1) and correctly substitute values.

4 Calculate the volume of gas present at 101.3 kPa and 25°C for each of the following.

a 8.7 moles

b 0.94 moles

c 2.5 moles

TRAP

Don't forget to use the correct formula (Figure 1) and correctly substitute values.

5 Calculate the molar mass of the following gas samples. All samples are at 101.3 kPa and 25°C.

a 11.4 g of a gas occupies 8.51 litres.

b 21.6 g of a gas occupies 6.89 litres.

c 4.6 g of a gas occupies 2.77 litres.

6 A student stated that when she raised the temperature of a gas from 20°C to 40°C, the temperature of the gas had doubled. Discuss this statement.

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8 CATALYSTS

Activation energy (E_A) is the energy required for a reaction to proceed. Activation energy is an important concept when describing how catalysts work. For particles to react with each other, they must collide with each other for the outer electrons to make contact and for bonding to occur.

1 What is the name for the outermost electrons of an atom?

In order for reactions to happen, bonds must be broken and the repulsion between electrons overcome. These processes require energy, and this energy is called activation energy. Different reactions have different values for the activation energy.

2 A reaction takes place spontaneously at room temperature.
What does this tell you about the activation energy for this particular reaction?

3 Think of reactions you have carried out that needed an input of energy before they would start.
Name two reactions and outline how energy was used to 'start them off'.

Figure 1 shows the reaction profile for an **exothermic** reaction. In these reactions, the energy level of the products is lower than that of the reactants, the enthalpy change (ΔH) is negative and heat is given off. In an **endothermic** reaction, the energy level of the products is higher than that of the reactants, the enthalpy change (ΔH) is positive and heat is absorbed.

4 Refer to figure 1 and then draw and label the equivalent diagram for an endothermic reaction.

TRICK Consider endothermic and exothermic reactions as 'mirror images' of each other.

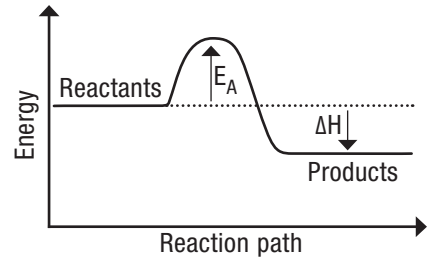

Figure 1

Catalysts speed up the rates of reaction without being consumed (used up) at the end of the reaction. They do this by altering the path of the reaction and lowering the activation energy required. The overall enthalpy change (ΔH) remains the same, only the activation energy (E_A) is lowered.

5 Copy figure 2 and label the catalysed and uncatalysed pathways.

HINT It helps to label the activation energies for the two pathways.

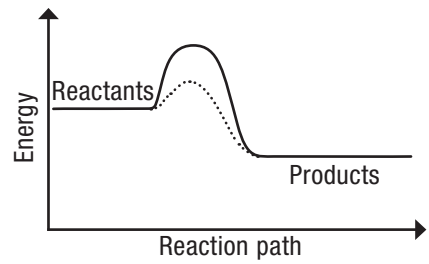

Figure 2

The decomposition of liquid hydrogen peroxide (H_2O_2) into water and oxygen is very slow at room temperature but speeds up considerably when a catalyst (MnO_2) is added.

6 Write a balanced equation for the decomposition of hydrogen peroxide.

7 What is the name of the chemical with the formula MnO_2?

TRAP Don't forget to include symbols of state and remember that oxygen is diatomic.

Catalysts are widely used in the chemical industry because they can lower the temperatures and pressures required and speed up reactions. This reduces energy and other costs. Many processes would not be economically viable if catalysts weren't used.

CATALYSTS AND THE CHEMICAL INDUSTRY

8 What is meant by *economically viable*?

TRICK The language used in Chemistry can be complex. If you have a good dictionary available, use it to help you 'translate' the more difficult terms into something you can understand easily.

If long chain alkanes are heated strongly enough in the absence of air, their molecules can vibrate so strongly that bonds are broken and smaller molecules are formed. This process is called *cracking*. At least one of the products is an alkene and catalysts are used to speed up the reactions. Figure 3 shows a typical catalytic-cracking reaction.

$$C_{11}H_{24} \rightarrow C_9H_{20} + H_2C = CH_2$$

undecane nonane

Figure 3

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9 Name the chemical with the formula CH_2CH_2.

10 Why is the catalytic cracking done in the absence of air?

THINGS TO KNOW Make sure you know your organic nomenclature.

The alkenes formed can be used to make other more complex molecules such as alkanols (alcohols) and polymers (plastics). Shorter chain hydrocarbons are often more valuable as fuels than longer chain hydrocarbons.

11 Give reasons for using a catalyst in the cracking process.

The **Haber Process** has been an important source of ammonia for a long period of time. Ammonia is an important raw material in the manufacture of nitrogen-based fertilisers. In an equilibrium reaction, ammonia gas (NH_3) is formed from hydrogen and nitrogen using an iron catalyst. Ammonia can be reacted with acids to give ammonium salts and it can be oxidised to form nitric acid.

12 Write a balanced equation for the formation of ammonia from hydrogen and nitrogen.

TRAP Don't forget that this is an equilibrium reaction and needs a special symbol between reactants and products.

13 Give the formula for the ammonium ion.

14 Give the formula for nitric acid.

The **Ostwald Process** is used to produce nitric acid from ammonia in a three stage process. The first stage is the reaction of ammonia with oxygen to form nitrogen monoxide and water using a platinum alloy catalyst.

15 Write a balanced equation for the reaction described above.

TRICK Don't try to write and balance an equation in one step. First write the equation in words. Then substitute formulae, balance the numbers and finally include symbols of state, i.e. (s), (l), (g) and (aq).

16 In the second stage, nitrogen monoxide is reacted with oxygen to give nitrogen dioxide. Write a balanced equation for this reaction.

TRICK When you see 'mono' in a formula, it means that there is one atom of the following element; 'di' means there are two atoms of that element and 'tri' means three atoms.

17 The final stage of the Ostwald Process involves the further oxidation of nitrogen dioxide and reacting with water. Balance the following reaction which shows this.

$$NO_{2(g)} + O_{2(g)} + H_2O_{(l)} \rightarrow HNO_{3(aq)}$$

The **Contact Process** produces sulfuric acid, which is then used in the manufacture of fertilisers, paints, fibres and detergents. The industrial production of sulfuric acid is a multi-stage process. The first step in this process is the manufacture of sulphur dioxide gas. The sulphur is sprayed into the reaction chamber and burned.

18 Write a balanced equation for the reaction described above.

TRICK When you see 'burned', then oxygen is a reactant and an oxide is very likely to be a product.

The second step of the Contact Process involves reacting sulphur dioxide with oxygen using vanadium (V) oxide as the catalyst to form sulphur trioxide gas.

19 Write a balanced equation for the reaction described above.

20 Give the formula for vanadium (V) oxide.

In the third step, the sulfur trioxide formed is reacted with water to form sulfuric acid (H_2SO_4).

21 Write a balanced equation for this reaction.

22 Decide whether each statement is true or false.

a Catalysts in a different physical state to the reagents are called heterogeneous catalysts.

b Solid catalysts provide a surface on which the reactant molecules temporarily stick in the correct orientation for an easier reaction.

TRICK Two key words here are *heterogeneous* and *orientation*. If you aren't familiar with their meanings, research them and add them to your glossary.

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9 WATER

Water has many special properties, both physical and chemical, which make it vital for life to exist and for many chemical reactions to occur. Water has been used to define units of measurement. It is no accident that under standard conditions, 1 mL (cm^3) of water has a mass of 1 g and that it freezes at 0°C and boils at 100°C.

Because water has such a high specific heat (4.18 Jg^{-1}), the large bodies of water on Earth minimise the changes in temperature between day and night. They absorb the heat from the Sun during the day and release it at night. Specific heat is the amount of heat (joules) required to raise the temperature of 1 g of a substance by 1°C.

TRAP Specific heat is a difficult concept. Make sure you can give a good definition and know how to apply it.

1 Describe three other physical properties of water.

2 How much energy would it take to raise the temperature of a cubic metre of water by 1°C? Show your working.

SLIP Don't forget to change volume to cm^3 (= mL).

The enthalpy of vapourisation is the energy required to transform a given quantity of a substance into a gas. Water has a very high enthalpy of vapourisation (heat of evaporation). Approximately 30% of the solar energy reaching the Earth's surface is used up in evaporating water from open bodies of water.

TRAP Don't confuse specific heat with enthalpy of vapourisation.

3 The human body uses a similar process to cool itself. What is the name of this process?

Most substances contract and become more dense when cold, and expand and become less dense when heated.

4 Using the previous facts and figure 1 to help you, copy and complete the following passage. Use these words: *convection, cycle, dense, displaced, oxygen, surface*.

In bodies of water, _____ plays a major part in circulating nutrients and _____. On cold nights, water on the _____ cools down and becomes more _____ and sinks to the bottom. The _____ water is pushed to the surface. The _____ continues with temperature changes at the surface.

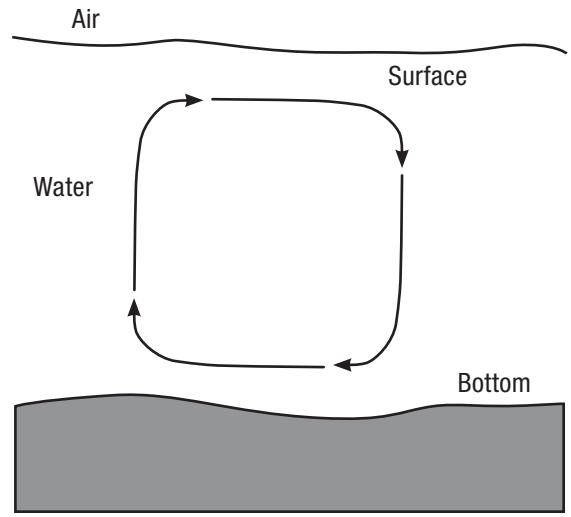

Figure 1

Water displays atypical properties as it approaches its freezing point. At around 4°C water reaches its maximum density. As the temperature falls below this point, instead of the density increasing and volume decreasing, water starts to expand. Cell walls in plants and animals can burst when subject to freezing. Because ice is less dense than liquid water, it floats on the surfaces of rivers and lakes, enabling life to continue below the surface.

5 What is meant by *atypical*?

6 What is the density of water under standard conditions?

7 Explain why water pipes can burst in winter.

TRAP Don't confuse *density* and *heaviness*. Density is mass present per unit of volume, i.e. mass divided by volume.

8 Decide whether each statement is true or false.

a Around 60% of Earth's surface is covered in water.

b *Aquatic* refers to water.

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10 STRUCTURE AND BONDING OF WATER

The unique properties of water are due to its bonding and structure. Figure 1 shows four different ways of representing the water molecule.

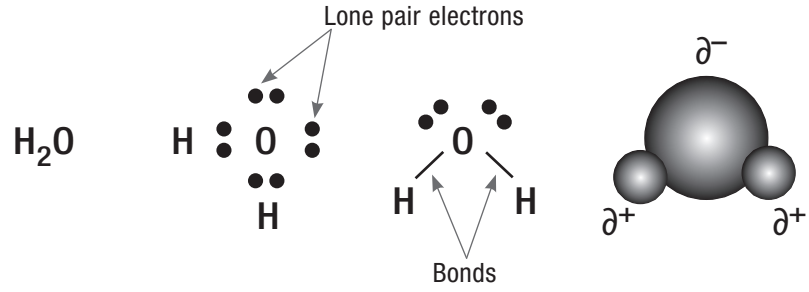

Figure 1

1 The bonds between hydrogen and oxygen are formed by sharing electrons. What type of bonding is this?

THINGS TO KNOW The major types of bonding include *ionic*, *covalent*, *polar covalent*, *hydrogen* and *Van der Waals (dispersion forces)*. Make sure you can identify each type and give a brief definition of them.

The water molecule is small and highly polar. The symbol δ^+ shows that one part of the molecule (hydrogen) is relatively positive and δ^- is relatively negative (oxygen). Water can dissolve many different ionic solids and polar covalent substances because bonds can be formed between the poles of water molecules and oppositely charged species.

TRAP Many students confuse polar molecules and ions. Polar molecules have some degree of charge separation – look for the delta (δ) symbol and a plus at one end of the molecule and delta symbol and a minus at the other end. Ions have either a complete positive or negative charge.

2 Figure 2 shows the interactions between water and a positive ion. Draw a similar diagram that shows how water molecules would interact with a negative ion.

HINT Opposites attract.

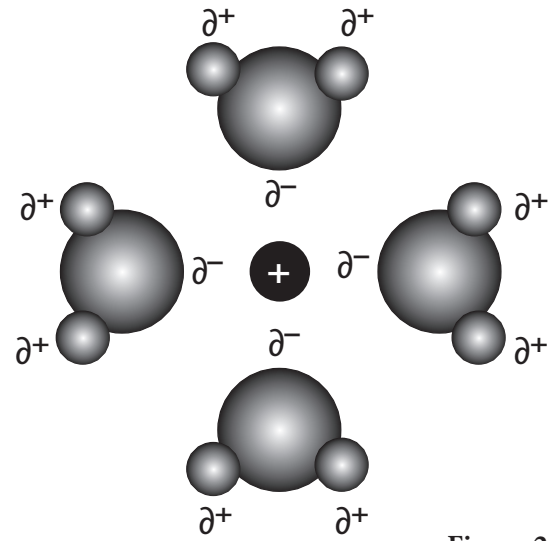

Figure 2

When ionic compounds dissolve in water they **dissociate** – the water molecules pull apart the ionic lattice forming hydrated ions. The energy released in forming the bonds between water and the ions is in the same order of magnitude as the energy needed to break down an ionic lattice. The positive ions are called cations and the negative ions are anions.

When sodium chloride dissolves in water, hydrated sodium ions and hydrated chloride ions are formed:

$$NaCl_{(s)} + H_2O_{(l)} \rightarrow Na^+_{(aq)} + Cl^-_{(aq)}$$

A hydrated ion has a specific number of water molecules connected to it.

3 Write equations for the dissociation in water of the following ionic compounds.

a sodium hydroxide, NaOH

b potassium nitrate, KNO_3

c sodium sulfate, Na_2SO_4

d ammonium chloride, NH_4Cl

e potassium phosphate, K_3PO_4

SLIP Be careful to ensure that you assign the correct charges to the ions formed and use these to help you balance the equations.

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Pure water is a very poor conductor of electricity, but when charged particles are present its conductivity increases enormously. Substances that dissolve in water, form ions and cause a current to flow are called electrolytes.

4 Complete the following passage using these words:
conductivity, dissociation, dissolve, electrical, hydrated, ions, polar.

Water molecules are _____, i.e. have parts which carry different _____ charges. When ionic compounds are _____ in water, they split into oppositely charged _____. This process is called _____. An ion that is surrounded by water molecules is said to be _____. When an electrolyte is present in water the _____ increases.

Substances that break down completely and cause a solution to conduct electricity very well are strong electrolytes. Ionic solids that dissolve tend to be strong electrolytes. Polar covalent molecules can also break down into ions when they dissolve in water in a process that is usually referred to as ionisation.

Hydrochloric acid (HCl), sulfuric acid (H_2SO_4) and nitric acid (HNO_3) are strong electrolytes (and hence strong acids). Hydrofluoric acid (HF) and ethanoic acid (CH_3COOH) are weak electrolytes (and hence weak acids). Covalent substances which remain as molecules when they dissolve are non-electrolytes. These include ethanol (C_2H_5OH) and sucrose ($C_{12}H_{22}O_{11}$).

5 Give a definition of a weak electrolyte.

THINGS TO KNOW
When you come across a new term, ensure that you have a concise definition for it. For example: An electrolyte is a chemical compound that ionises when dissolved to produce an electrically conductive medium.

The greater the concentration of an electrolyte, the greater the conductivity of the solution. The concentration of a solution of a known electrolyte can be found using a conductivity versus concentration graph.

6 The following table shows the conductivity for a range of solutions of sodium chloride (NaCl). Draw a scatter graph, with concentration on the *x*-axis and conductivity on the *y*-axis. Then draw a line of best fit.

TRAP
Label the axes and include units. Don't just 'join the dots' for the line of best fit – make sure you draw a smooth curve.

Conductivity for sodium chloride (NaCl) solutions						
Concentration (moles L^{-1})	0	0.2	0.4	0.6	0.8	1.0
Conductivity (arbitrary units)	0	2.1	3.9	5.5	7.1	8.5

7 What would be the concentration of a solution of sodium chloride that had a conductivity of 6.3 arbitrary units?

HINT
Draw a line from 6.3 on the *y*-axis until it hits the curve. From this point, draw a line down until it hits the *x*-axis.

8 How would the conductivity of similar solutions of magnesium chloride ($MgCl_2$) compare with those of sodium chloride? Explain your answer.

HINT
Think about the ions that these two compounds would form.

9 Give the names and formulae of two polar covalent compounds that are strong electrolytes.

10 Give the names and formulae of two polar covalent compounds that are weak electrolytes.

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11 WATER QUALITY

Water is used in large quantities by industry, agriculture, businesses and homes. As well as being an essential resource for humans, water is the habitat (home) for many organisms. Whenever water is used by humans, it is almost inevitable that the water will be polluted by the human use.

1 Name two types of organisms that live in water.

2 Give a definition for *pollution*.

Water has to be taken from where it exists to where it is needed. Water can be pumped from a river, dam or bore.

3 Outline one other way that people can collect and distribute water.

THINGS TO KNOW

When you come across a new term, learn a concise definition for it. For example: *Reticulated water is water supplied by a network of pipes.*

Most urban and city areas have reticulated ('town') water. This has been collected in a catchment area, stored in a dam and treated before being piped to the users.

4 What is a catchment area?

5 Copy and complete the following passage using these words:
aluminium, carbon, chlorine, diseases, filter, fine, purifying, sedimentation.

The process of _____ water involves several steps. Suspended material is allowed to settle out in a process called _____. Very _____ suspended material is removed by the addition of _____ ions (Al^{3+}) which form a precipitate. Then the water is passed through a _____ to screen out any remaining solids. Bacteria can cause _____ and are usually killed by the addition of _____ (Cl_2) or ozone (O_3). Any residual taste or odours can be removed by passing through activated charcoal (_____, C).

In many areas of the world there is a shortage of potable (safe to drink) water. One way to obtain fresh water is to remove the salts from sea water or brackish (slightly salty) water. The two most common large scale processes used are **distillation** and **reverse osmosis**.

6 The stages of distillation are shown below, but not in the correct sequence. Rearrange them into the correct order.

a Salt is left behind
b Salt water enters the system
c Salt water is heated
d Steam condenses to form pure water
e Water boils and turns to steam

Distillation does have disadvantages. Large amounts of energy are required to boil the water which makes the process expensive and generates large quantities of Greenhouse gases.

Figure 1 shows how water can be purified by reverse osmosis. Fresh water is separated from salty water by a porous membrane. If pressure is applied to the salty water, water moves from the solution through the membranes to the pure water. Reverse osmosis uses about one third of the energy needed for distillation per litre produced.

7 State other drawbacks of distilling water on a large scale.

8 Copy figure 1 and add the following labels.

a membrane
b pressure
c pure water
d salty water.

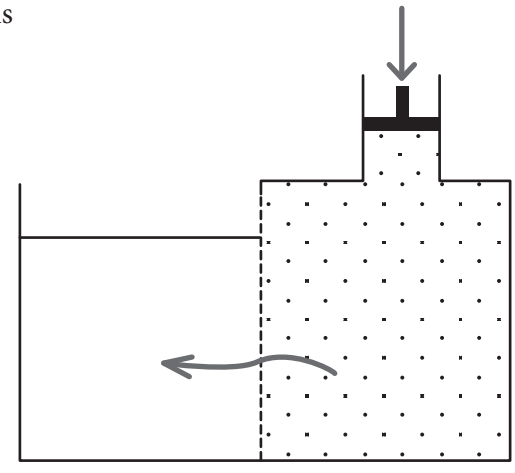

Figure 1

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12 WATER POLLUTION

Polluted water is water that is not fit for the purpose for which it is intended. Pollutants in water can include toxic chemicals and pathogens. When we refer to 'water pollution', we tend to concentrate on contamination that has been caused by human activity.

The manufacturing industry is responsible for a major part of water pollution. Many factories discharge polluted water (effluent) into rivers, lakes or oceans. Although there are statutory limits on the amounts of pollutants that can be in this waste water, total volumes can be huge and can severely degrade water quality. Industrial pollutants that enter water supplies include heavy metals (e.g. mercury), organic solvents and oils.

THINGS TO KNOW

When you come across an important term, ensure that you have a concise definition for it. For example: Pollution is a change in the environment that is harmful to its existing or potential uses.

1 Name a manufacturing industry that contributes to water pollution. Describe the type of pollution it generates and the possible impact of this pollution.

2 The building industry contributes surface run-off, such as sediment from building sites. Name another aspect of the building industry that contributes to water pollution. Describe the possible impact of this pollution.

HINT

Think about the materials used on a building site.

Generating large quantities of electricity always impacts on the environment. Power stations can burn fossil fuels (coal, oil or natural gas), use nuclear power or utilise moving water or wind. When coal is burned, it forms sulfur dioxide (SO_2) which is a major contributor to acid rain. When acid rain falls it enters bodies of water, causing them to become too acidic for aquatic life. Both coal-fired power stations and nuclear power stations require large amounts of water for cooling. The hot water discharged into rivers or oceans can cause thermal pollution.

3 Discuss the possible consequences of thermal pollution in a body of water.

4 Discuss another example of water pollution caused by electricity generation that has not already been mentioned.

HINT

Consider living things present in water.

Households discharge water containing detergents, sewage and household chemicals. This waste is usually treated before it is released into waterways. However, treatment doesn't remove phosphorus, which stimulates the growth of certain aquatic organisms. This in turn results in the level of dissolved oxygen decreasing, which is called eutrophication.

5 Discuss the effects of eutrophication.

6 Give two other examples of water pollution caused by households.

THINGS TO KNOW

When you come across a new term, ensure that you have a concise definition for it. For example: Eutrophication is the undesirable overgrowth of vegetation caused by high concentrates of plant nutrients in bodies of water.

Modern agriculture uses many chemical inputs such as pesticides and fertilisers containing nitrates and phosphates. Excess quantities of these substances tend to be carried into water ways by rain ('run-off'). Some substances degrade quickly and others take longer periods and can be absorbed by aquatic organisms.

7 Discuss a different example of water pollution caused by agriculture.

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13 SOLUTIONS

When we add a solute (gas, liquid or solid) to a solvent (usually a liquid) and the solute particles disappear, we say that a solution has been formed. Most solutions you will deal with tend to be *aqueous*, i.e. water is the solvent.

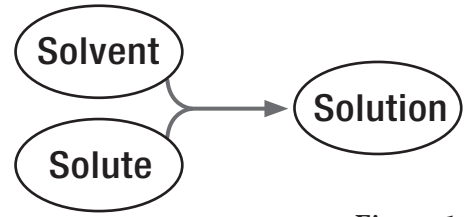

Figure 1

HINT If you see the word 'aqueous' it means *contains, is connected to or is dissolved in water*.

TRAP Many students confuse the terms solvent, solute and solution. Figure 1 shows the relationship between these three terms.

1 Describe three ways to speed up the dissolving of a solid in water.

There is a great deal of variety in how well different substances dissolve in water. Ethanol (methylated spirits) is soluble in all proportions (i.e. *miscible*) with water. Others such as silver chloride are either very poorly soluble or insoluble in water.

2 What does *soluble in all proportions* mean?

Solutions are defined as homogenous mixtures; they have particles uniformly spread through the mixture. Although solutions will pass through filter paper, like most mixtures, the individual components in a solution can be separated.

3 Outline how you could recover the salt from an aqueous solution of common salt.

4 Outline how you could recover the water from an aqueous solution of common salt.

THINGS TO KNOW If you find these two questions difficult, revise your work on separation of substances.

Water is a polar molecule, as illustrated in figure 2. This means that one end is relatively electrically charged compared to the other end. Organic (carbon-based) solvents tend to be slightly polar (e.g. ethanol, figure 3) or non-polar (e.g. benzene, figure 4).

5 Give the molecular formula for:

a ethanol.

b benzene.

HINT The molecular formula shows the numbers and kinds of atoms in a molecule.

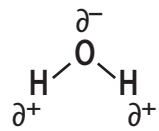

Figure 2

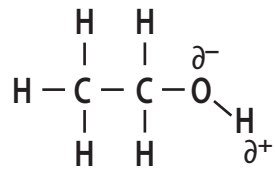

Figure 3

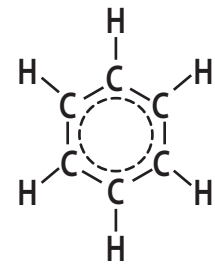

Figure 4

The phrase, 'Like dissolves like' is often used to predict solubility. The basis behind this 'rule of thumb' is that a solute will dissolve best in a solvent that has a similar polarity. This is not a definitive rule, but it can be useful in forecasting solubilities. For a substance to dissolve, the particles making up the substance need to move away from each other and spread throughout the solvent. The type and strength of bonding in the solute and solvent will affect whether this can happen. The bonds between individual particles of the solute must break (energy supplied) before they can form new interactions with water (energy given out).

6 Copy and complete this sentence: A _____ solute will dissolve _____ in a polar solvent and a non-polar _____ will dissolve best in a _____ solvent.

7 Give the general name for a reaction which:

a absorbs energy from its surroundings.

b releases energy to its surroundings.

TRICK *Thermic* refers to heat, *exo* means 'going out', *endo* means 'coming in'.

Covalent and ionic substances behave in different ways when they dissolve in water. Remember the two main types of covalent structures: covalent network (macromolecules) and molecular. Covalent network structures such as diamond and silicon dioxide usually don't dissolve in water or in other solvents.

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WATER WITH COVALENT AND IONIC SUBSTANCES

8 Give possible reasons for covalent network substances tending to be insoluble.

THINGS TO KNOW Make sure you remember theory of bonding. Revise ionic and covalent, covalent network, covalent molecular and polar covalent.

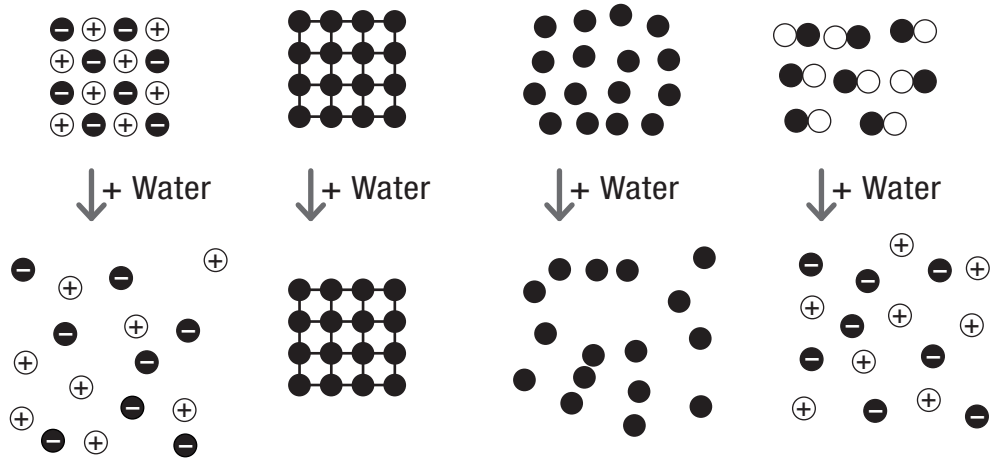
If covalent molecular substances do dissolve in water then the loosely bound structures tend to split into individual molecules.

The molecules tend to remain intact, but have interactions with water molecules. Polar covalent molecules (e.g. hydrogen chloride, HCl) can split into positive ions (cations) and negative ions (anions) which are attracted to the polar ends of the water molecule. If ionic substances do dissolve in water then the strongly bound structures (ionic lattices) tend to split into oppositely charged ions.

9 Give an example of a substance that will exist in an ionic lattice.

10 The diagrams at the right show how various types of compounds behave when they are added to water. Identify the type of bonding most likely present for each substance.

TRICK Ionic compounds tend to be formed between Group I and Group VII elements.



INTERACTIONS BETWEEN WATER AND IONS IN SOLUTION

When an ionic compound such as sodium chloride (NaCl) dissolves in water, the lattice dissociates into individual ions that are *solvated*, i.e. surrounded by water molecules. Sodium chloride dissolves in water due to the attractions between the polar water molecules and the sodium and chloride ions. Figure 6 shows the ion-dipole attraction between a chloride ion and a water molecule.

11 Referring to figure 5 as a model, use similar notation to show the interaction between a sodium ion and a water molecule.

HINT Opposites attract.

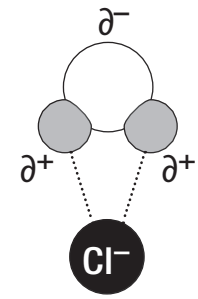


Figure 5

Rules of solubility in water	
Soluble compounds • Most Group I (alkali metals) and ammonium (NH_4^+) • Chlorides, bromides and iodides – exceptions are those containing silver, mercury or lead • Fluorides – exceptions are those with Group II (alkali earth metals) • All nitrates (NO_3^-) • Sulfates (SO_4^{2-}) – exceptions are those containing barium, calcium, silver, lead or mercury	**Insoluble compounds** • Carbonates (CO_3^{2-}) – exceptions are those containing Group I metals and ammonium. • Phosphates (PO_4^{3-}) – exceptions are those containing Group I metals and ammonium.

12 Draw up a table with the column headings: 'Name', 'Formula' and 'Solubility.' List the following compounds in the 'Formula' column. Then complete the 'Name' and 'Solubility' columns.

$NaNO_3$	$PbCl_2$	Li_2CO_3	$Ca_3(PO_4)_2$	$Ca(NO_3)_2$	NH_4Br
KCl	NH_4Cl	$HgSO_4$	BaI_2	$(NH_4)_2CO_3$	$Pb(NO_3)_2$
$MgSO_4$	CaF_2	$AgNO_3$	$MgBr_2$	$Li_3(PO_4)$	K_2CO_3

SLIP Consider both the metal and non-metal parts of these compounds when deciding if they are soluble.

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14 SATURATED SOLUTIONS

If we add portions of sugar to a fixed mass of water and stir to dissolve it, eventually there will be a point at which no more of the sugar will dissolve. At this point the solution is saturated.

THINGS TO KNOW

Saturation is an important concept. Ensure that you have a concise definition for it. For example: A saturated solution contains the maximum amount of a solute that will dissolve under stated conditions.

1 Copy and complete the sentences.

A _____ solution is one in which no more _____ will dissolve under the existing conditions of _____ and pressure. An _____ solution is able to dissolve more solute under the existing conditions.

The solubility of a solute in a particular solvent can be expressed qualitatively; e.g. high, slight, low. It can also be shown quantitatively as a concentration (e.g. grams of solute per 100 g of solvent or moles per litre). For almost all solutes, solubility in a given solvent will depend on temperature. This can be shown graphically in a solubility curve where solubility (*y*-axis) is plotted against temperature (*x*-axis). The line of a solubility curve shows the concentration of a saturated solution at specific temperatures.

The table below shows the solubilities of potassium nitrate and sodium chloride in water for a range of temperatures.

Temperature (°C)		**0**	**10**	**20**	**30**	**40**	**50**	**60**	**70**	**80**	**90**	**100**
Solubility (g/100 g)	KNO_3	14.1	23.2	33.0	46.1	62.0	80.8	102.3	128.9	Not measured		
	NaCl	36.0	36.5	37.0	37.5	38.0	38.5	39.0	39.5	40.0	40.5	41.0

2 Use the information in the table to draw solubility curves for the two solutes.

TRAP

Label the axes and include units. Don't just 'join the dots' for the line of best fit – make sure you draw a smooth curve.

3 Describe how the solubility of potassium nitrate changes with temperature.

4 Describe how the solubility of sodium chloride changes with temperature.

5 Use your graph to work out the amount of solute per 100 g of solvent for the following. Show how you obtained the answers.

a potassium nitrate at 65°C

b potassium nitrate at 25°C

c sodium chloride at 65°C

d sodium chloride at 25°C

HINT

Draw a line from temperature on *x*-axis until it hits the curve. From this point, draw a line across to the *y*-axis.

6 Forecast what would happen if a saturated solution of potassium nitrate at 65°C containing 100 g of solvent was cooled to 25°C.

SOLUBILITY OF GASES

The trend of solubility of gaseous solutes in water (Figure 1) and in organic solvents (Figure 2) is shown in the graphs.

7 Describe how the solubility of gases in water changes with temperature.

8 Describe how the solubility of gases in organic solvents changes with temperature.

TRICK

When looking at relationships shown by a graph, ask yourself, *What happens to y when x changes?*

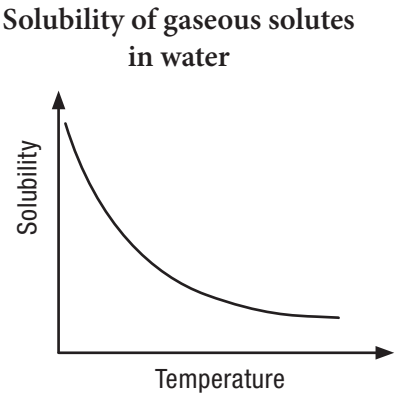

Figure 1

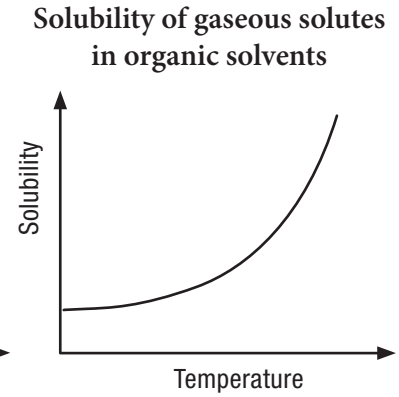

Figure 2

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9 A power station is being planned. It will discharge hot water into a body of water. Discuss the potential effects on aquatic life.

> **HINT** Thermal pollution

10 Referring to figure 3, describe how pressure affects the solubility of gases.

11 Fizzy drinks ('carbonated beverages') are bottled under pressure to increase the carbon dioxide dissolved in solution. Discuss what happens when a bottle of fizzy drink is opened.

12 **Research:** Find out why deep-sea divers can suffer a disease called 'the bends' (decompression sickness).

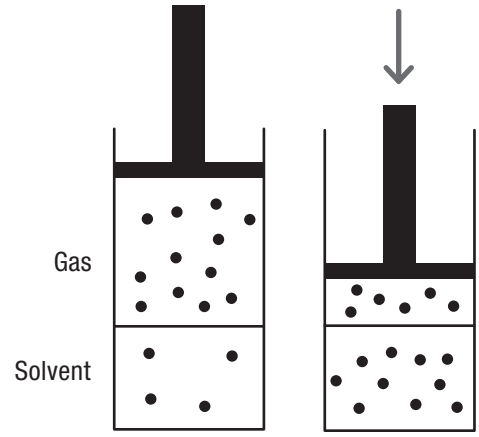

Figure 3

Hyperbaric chambers are pressurised vessels where patients can be treated for certain complaints. The patient enters the chamber and pressure is increased up to six times normal atmospheric pressure. Pure oxygen is often used.

13 Give possible effects on a patient of treatment in a hyperbaric chamber.

> **HINT** Consider what would happen to the patient's metabolism if the amount of oxygen taken in is increased considerably.

SOLID SOLUTIONS

Metals can be melted and dissolved with each other to form alloys such as brass (copper and zinc), solder (lead and tin) and steels (iron-based). Alloys can be designed to have more desirable properties than the separate metals of which they are made.

14 'Gold' coins in circulation in Australia are made of an alloy that contains no gold at all. It has 92% copper, 6% aluminium and 2% nickel. Give possible reasons for using this alloy for 'gold' coins.

> **HINT** Think of the physical properties of gold and its cost.

15 Pure gold is described as 24 carat. It is frequently alloyed with other metals such as silver and copper. A piece of jewellery is described as being 18 carat gold. Calculate the percentage of gold present and show your working.

16 Solder is used to permanently join electrical connections. It is heated and melts. As it cools and solidifies, it joins the parts. List two desirable properties of solder.

17 **Research:** Glass can be described as a 'super-cooled solution'. Find out what this means.

18 Match each type of solution with the correct example.

Types of solution	Examples
a Liquid in liquid	• Soda water
b Gas in liquid	• Wine
c Gas in gas	• Stainless steel
d Solid in solid	• Air

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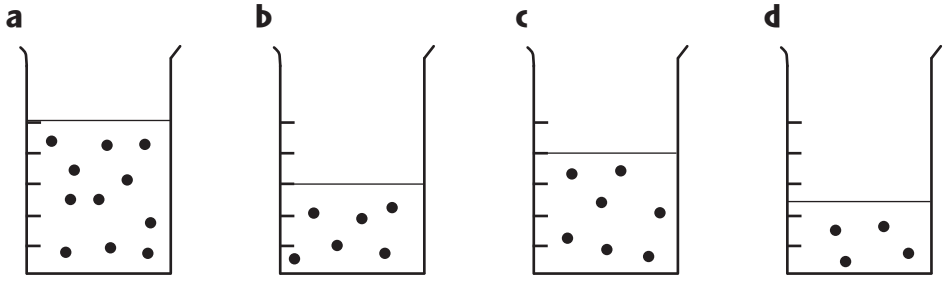
15 SOLUTIONS, CONCENTRATION AND STRENGTH

Some substances are soluble in water. Water is known as the **solvent**. The substance being dissolved is the **solute** and the resulting mixture is the **solution**. The terms used to describe solutions, i.e. *concentrated, dilute, strong* and *weak,* are often confused. Concentration tells us the quantity of a solute (substance that has been dissolved) present in a given volume of solution. Concentration is quoted as a ratio, e.g. grams per litre ($g.L^{-1}$) or moles per litre ($mol.L^{-1}$, M). A concentrated solution contains more solute than a dilute solution.

1 Figure 1 shows four 500 mL beakers, each containing a solution of different concentration. Arrange the four solutions in order of *increasing concentration*. Explain how you arrived at your answer.



HINT Remember that concentration is a ratio.

• Indicates relative number of particles of solute

Figure 1

2 Calculate the concentration of each solution in grams per litre.

a 5.0 g solute in 500 mL of solution

b 7.6 g solute in 200 mL of solution

c 41.3 g solute in 2 L of solution

d 19.2 g solute in 750 mL of solution

SLIP Don't forget to change volumes to litres.

To help work out the ratios involved when solutions react (stoichiometry), we have to be able to deal with molarities of solutions. See figure 2.

$$C = \frac{n}{v}$$

$$\text{Concentration} = \frac{\text{number of moles}}{\text{volume (litres)}}$$

Figure 2

3 Calculate the concentration of the each solution in moles per litre.

a 0.5 moles of solute in 200 mL of solution

b 0.25 moles of solute in 500 mL of solution

c 1.3 moles of solute in 1500 mL of solution

d 0.1 moles of solute in 250 mL of solution

When substances dissolve in water, they can stay as discrete (separate) molecules or they can dissociate (break up) into positive and negative aqueous ions. If the solute is an ionic compound then all of the solute dissociates. For example, if sodium chloride dissolves then this can be represented by the equation:

$$NaCl_{(s)} + H_2O_{(l)} \rightarrow Na^+_{(aq)} + Cl^-_{(aq)}$$

4 Write an equation for the reaction when sodium hydroxide (NaOH) dissolves in water.

Soluble molecular covalent substances tend to remain as molecules when they dissolve. For example, when glucose dissolves this can be represented by the equation:

$$C_6H_{12}O_{6(s)} + H_2O_{(l)} \rightarrow C_6H_{12}O_{6(aq)}$$

Polar covalent substances have bonding intermediate between ion and molecular covalent. One end of the molecule is slightly more positive than the other. When a polar covalent molecule dissolves, some of the molecules break up into positive ions and negative ions. See figure 3.

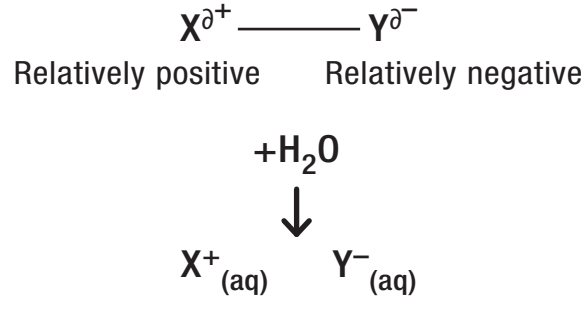


Figure 3

5 Give a definition for *polar covalent*.

TRICK Try to use a source other than your usual textbook to find definitions.

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Polar covalent substances can dissociate completely (strong electrolytes) or in only negligible amounts (weak electrolyte). These are equilibrium reactions and can be shown as:

$$XY_{(s)} + H_2O_{(l)} \rightleftharpoons X^+_{(aq)} + Y^-_{(aq)}$$

In strong electrolytes, the position of equilibrium lies to the right; i.e. almost all of the species present are the products.

6 In a weak electrolyte, what is the ratio between reactants and products?

Figure 4 shows the behaviour of two different polar covalent substances (X and Y) when they are added to water.

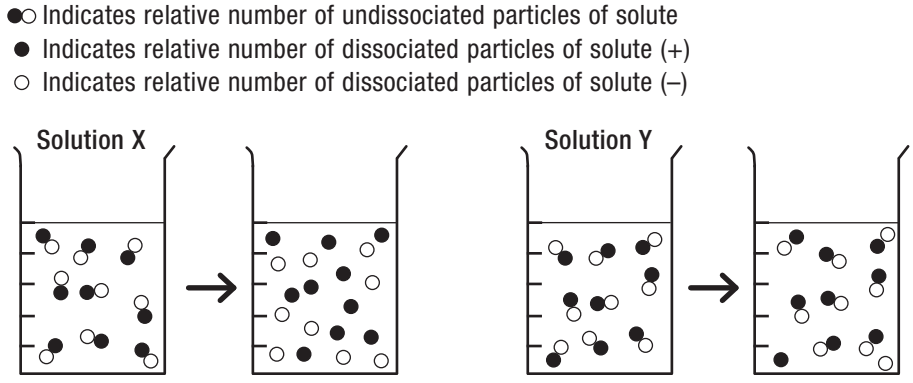

Figure 4

7 Identify which substance in figure 3 is a strong electrolyte and which is a weak electrolytes. Justify your answers.

HINT Strong electrolytes dissociate almost completely. Weak electrolytes only dissociate slightly.

The concept of strength is very useful when describing acids. The dissociation of an acid can be shown as:

$$HA_{(aq)} + H_2O_{(l)} \rightleftharpoons H^+_{(aq)} + A^-_{(aq)}$$

In a strong acid, the position of equilibrium lies to the right; i.e. almost all of the species present are the products. Strong acids include hydrochloric (HCl), nitric (HNO_3) and sulfuric (H_2SO_4) acids.

TRAP Many students confuse the terms *strong and weak* with *concentrated and dilute*.

8 Write formula equations showing the dissociation of the following acids when they dissolve in water. (Assume that sulfuric acid dissociates to give two hydrogen ions per molecule.)

SLIP Don't forget that these are equilibrium reactions.

a Hydrochloric acid **b** Nitric acid **c** Sulfuric acid

In a weak acid the position of equilibrium lies to the left, i.e. almost all of the species present are the reactants. Weak acids include methanoic (CHOOH), ethanoic (CH_3COOH), carbonic (H_2CO_3) and phosphoric (H_3PO_4) acids.

9 Write formula equations showing the dissociation of the following acids when they dissolve in water. (Assume that carbonic acid dissociates to give one hydrogen ion per molecule, and that phosphoric acid dissociates to give three hydrogen ions per molecule.)

a Methanoic acid **b** Ethanoic acid

c Carbonic acid **d** Phosphoric acid

Bases can also be classified as strong or weak. Sodium hydroxide (NaOH) and potassium hydroxide (KOH) are the major strong bases. Ammonium hydroxide (NH_4OH) and calcium hydroxide ($Ca(OH)_2$) are common weak bases.

10 Write formula equations showing the dissociation of the following bases when they dissolve in water. (Assume that calcium hydroxide dissociates to give two hydroxide ions per molecule.)

SLIP Be careful balancing the calcium hydroxide equation.

a Ammonium hydroxide **b** Potassium hydroxide **c** Calcium hydroxide

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16 THE EARTH'S RESOURCES

A resource is anything we use to help maintain our lifestyles. These resources can be naturally occurring or synthetic. They can be renewable or non-renewable (finite). Resources are obtained from the biosphere, lithosphere, hydrosphere and atmosphere.

1 Match each term with the correct meaning.

Terms	Meanings
a Atmosphere	• Solid outer layer of the Earth
b Biosphere	• Layer of gas surrounding the Earth
c Finite	• Part of the Earth where water is present
d Hydrosphere	• Will eventually run out
e Lithosphere	• Made by humans
f Synthetic	• Zone where life exists

HINT If you have trouble with the meanings of these words, make sure you include them in a glossary.

2 Draw up a table like the one below. Then study the list of substances. Copy them into the correct column of your table to show where they occur in nature. Note that some will occur in more than one area. Make a separate note of any substances which you cannot identify.

Substances

- argon
- sand
- oxygen
- water vapour
- blood
- humus
- sodium chloride
- wood
- water
- helium
- blood
- clay
- coal
- nitrogen
- gypsum
- hydrocarbons
- orange juice
- soot
- pollen
- granite
- cytoplasm
- iron ore
- soil
- plankton
- calcium carbonate
- nitrogen oxides
- gold
- bauxite
- neon
- limestone
- ice
- glucose
- carbon monoxide
- marble
- sulfur
- carbon
- sulfur dioxide
- sap

Biosphere	Lithosphere	Hydrosphere	Atmosphere

Mining is an extremely important industry and supplies the raw materials that provide us with energy and manufactured items. Metals play a significant part in our lifestyles. However, metals rarely occur uncombined (native) in the Earth's crust – they are more often in the form of oxides, carbonates, chlorides, silicates or sulphides.

3 Name a metal that can be found uncombined in the Earth's crust.

HINT Consider reactivity.

4 Name and give the symbols for the non-metallic elements present in:

a oxides.

b carbonates.

c chlorides.

d silicates.

e sulfides.

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17 EXTRACTION OF METALS

1 Copy and complete the following passage, using these words: *deposit, impurities, occurring, reserve, useful.*

A mineral is a naturally _____, inorganic solid with a specific chemical composition and characteristic crystal structure. An ore is a material from Earth's crust which contains enough of a _____ mineral to be worth mining. A _____ is where an ore has been laid down or accumulated. The _____ of an ore is the amount known to exist that can be extracted. _____ have to be removed before metals can be obtained from their ores, often by smelting.

Industrially, metals tend to be extracted from their ores by simply heating, by reduction with carbon or by electrolysis. The method used depends on the reactivity of the metal.

THINGS TO KNOW Make sure you have a working definition for reduction. For example: *Reduction is the removal of oxygen and/or gain of electrons.*

Smelting is the heating of an ore to produce the metal. Lead and copper are relatively unreactive and heating their ores will produce the (impure) metals.

2 If lead sulfide (galena, PbS) is heated in air, lead and sulfur dioxide are produced. Write a balanced equation for this reaction.

SLIP Where does the oxygen come from in sulfur dioxide?

3 If copper sulfide (Cu_2S) is heated in air, copper and sulfur dioxide are produced. Write a balanced equation for this reaction.

SLIP Don't forget symbols of state.

Tin, iron and zinc ores are reduced using carbon.

4 When tin (IV) oxide (SnO_2) is heated with carbon, tin and carbon dioxide are formed. Write a balanced equation for this reaction.

The extraction of iron from iron (III) oxide (haematite, Fe_2O_3) is a little more complex. Carbon monoxide (CO) formed in the furnace reacts with haematite to form iron and carbon dioxide.

5 Write a balanced equation for the reaction between carbon monoxide and iron (III) oxide.

SLIP Watch how you balance this equation.

It is not practical to reduce the oxides of the more reactive metals (potassium, sodium, calcium, magnesium and aluminium) by heating with carbon or carbon monoxide. Instead, electricity is used to split the compound. Producing metals by electrolysis is expensive due to the huge amounts of energy involved. The ore has to be fused (molten) to allow ions to move and a current to flow.

In an electrolytic cell:

- the anode is positive
- oxidation takes place at the anode
- the cathode is negative
- reduction takes place at the anode.

THINGS TO KNOW Make sure you have a working definition for electrolysis. For example: *Electrolysis is a chemical change produced in a liquid by an electric current. Oxidation is the addition of oxygen and/or loss of electrons.*

Aluminium is produced on a large scale by electrolysis from alumina (Al_2O_3).
The aluminium oxide is molten – cryolite (Na_3AlF_6) is added to lower the melting point.

6 At the anode, the oxygen ions lose electrons and are evolved as oxygen gas. Write a balanced equation for this reaction.

7 At the cathode, the aluminium ions gains electrons and are discharged as the liquid metal. Write a balanced equation for this reaction.

SLIP Make sure you balance the number of electrons and charges on ions.

8 Sodium can be obtained from sodium chloride using a method similar to the production of aluminium.

a Write a balanced equation for the reaction at the anode.

b Write a balanced equation for the reaction at the cathode.

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18 TRANSITION METALS

Transition metals occupy the middle of the Periodic Table, between Groups II and III. They are sometimes known as the 'd-block' elements. Most definitions classify transition metals as those with incomplete inner d-subshells. They don't follow the normal way of electrons filling shells in sequence. Copper, cadmium and zinc have complete d-subshells and so don't meet this criterion. Copper and cadmium are often included despite this, because they have physical and chemical properties in common with the main body of transition metals.

Sc		Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Cu
Y		Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
La	*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg

*Lanthanides

1 Outline how electrons fill up shells for the first 20 elements.

HINT Remember the 'octet' rule.

As you move left to right across the Periodic Table, generally periodic trends appear. These include changes in ionisation energy, increasing metallic character and decreasing ionic radius. The transition metals have fewer differences between themselves than most other neighbouring elements. For example, they have more than one oxidation state and transition metal compounds are often coloured.

THINGS TO KNOW Make sure you know the general features of the Periodic Table and how it is organised.

2 Explain what is meant by *ionisation energy*.

THINGS TO KNOW Ionisation energy is an important concept. If you have trouble with this question, spend some time revising the concept.

3 Give three physical properties of metals.

4 Give three chemical properties of metals.

5 What are the lanthanides?

Transition metals tend to have the following properties:

- strong
- can act as catalysts
- form several cations
- high melting and boiling points.

6 What is a *cation*?

7 Give two other properties of transition metals.

Any atom that is involved in ionic or covalent bonding can be assigned an oxidation number. This number tells us whether the atom has gained or lost control of electrons and also the number of electrons involved. The oxidation number of an atom in a pure element is zero. If the oxidation number is negative, then the atom has gained electrons; if it is positive, it has lost electrons. For example, Mn (VII) is manganese with an oxidation number of +7; that is, it has lost 7 electrons.

Because transition metals have variable oxidation states, the oxidation number is often shown in Roman numerals in brackets as part of the formula; e.g. copper (II) sulfate and iron (III) oxide. The oxidation state (number) of an atom shows how control of electrons has been transferred to or from that atom as a result of bonding.

8 Give the formulae for the following compounds.

a potassium manganate (VII)

b chromium (III) oxide

c iron (II) hydroxide

d cobalt (II) nitrate

TRAP The (VII) refers to manganese. This substance is commonly called potassium permanganate.

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9 Refer to table 1 to answer the following questions.

a Are the values I to VII positive or negative? Explain your answer.

TRICK Metals tend to lose electrons and form positive ions.

b Which metal has the greatest number of stable oxidation states?

c Sometimes scandium is not considered to be a transition metal. Give a reason for this.

Table 1: The range of stable oxidation states for some transition metals

Stable oxidation state	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu
I									✔
II		✔	✔	✔	✔	✔	✔	✔	✔
III	✔	✔	✔	✔	✔	✔	✔		
IV		✔	✔		✔			✔	
V			✔						
VI				✔	✔	✔			
VII					✔				

The ions of transition metals often have distinctive colours, as opposed to Group I and II metals which tend to have white compounds and colourless solutions. Table 2 shows the colours of aqueous solutions of some transition metal ions.

Table 2: The colours of aqueous solutions of some transition metal ions

Stable oxidation state	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu
I									
II					pink	green	pink	green	blue
III		violet	blue	green		yellow	yellow		
IV			blue						
V			orange						
VI				orange	green	violet			
VII					purple				

10 What are the possible transition metal ions present in solutions that have the following colours? Assume that the colour is not caused by a mixture of ions.
Include the name, symbol, charge and oxidation state in each answer.

a green **c** yellow **e** orange **g** blue

b pink **d** purple **f** violet

Because of their variable oxidation states, transition metals undergo many reduction/oxidation (redox) reactions. An atom that loses electrons has been oxidised. An atom that gains electrons has been reduced.

11 In each reaction, identify whether the initial ion has been oxidised or reduced.

a $Fe^{2+}_{(aq)} \rightarrow Fe^{3+}_{(aq)} + e^-$

b $V^{5+}_{(aq)} + 2e^- \rightarrow V^{3+}_{(aq)}$

c $Mn^{2+}(aq) \rightarrow Mn^{7+}_{(aq)} + 5e^-$

d $Co^{2+}_{(aq)} \rightarrow Co^{3+}_{(aq)} + e^-$

e $Cu^{2+}_{(aq)} + 2e^- \rightarrow Cu_{(s)}$

THINGS TO KNOW Make sure you know what a change in oxidation state means in term of reduction and oxidation.

12 What observation would tell you that the reactions in question 11 had occurred?

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19 CHROMATOGRAPHY

Chromatography is the process of separating small amounts of substances from mixtures by the relative speeds at which they move through or along a medium. Chromatography was first used by Mikhail Tsvet in 1901 to separate plant pigments.

1 Give another term for *pigment*.

There are many chromatographic techniques, but they all rely on the components of a mixture being carried by a mobile phase over (or through) a stationary phase (the medium) at different rates. The molecules in the mixture will have different interactions with the stationary phase. The greater the interaction, the slower the substance will move through the stationary phase.

Chromatography is used to identify dyes, additives, pharmaceuticals and many other organic compounds. Chromatography is especially useful in forensic chemistry.

2 What is *forensic chemistry*?

Paper chromatography uses a liquid solvent as the mobile phase and paper as the stationary phase – strictly speaking, it is water absorbed by the paper that is the stationary phase. The mixture to be separated is dissolved in the minimum amount of a solvent. This is placed on the surface of the paper. A second solvent flows across the sample, carrying it through the paper. (See figure 1.) Radial chromatography and ascending chromatography are the two most common examples.

Figure 2 shows the sequence followed in a simple radial chromatography procedure.

TRAP

A common misconception is that chromatography can be used to separate and identify only coloured compounds. You will be familiar with separating the colours in inks and grass, but chromatography can also be used with mixtures that don't contain coloured substances.

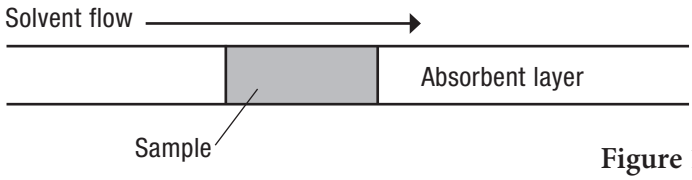

Figure 1

THINGS TO KNOW

Make sure you have a clear understanding of important chemical processes such as **chromatography** and relevant key terms such as *stationary phase* and *mobile phase*.

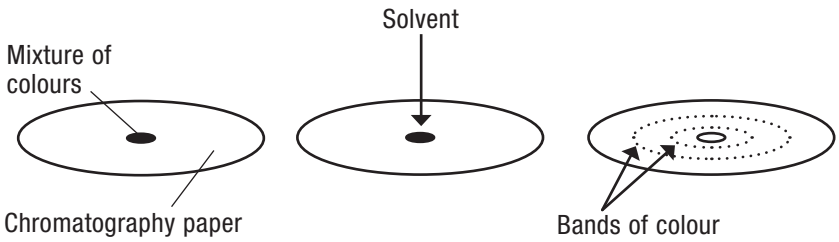

Figure 2

3 Outline how radial chromatography is carried out.

Figure 3 shows a 'before and after' example to illustrate ascending chromatography.

4 Referring to figure 3:

a identify the stationary phase.

b identify the mobile phase.

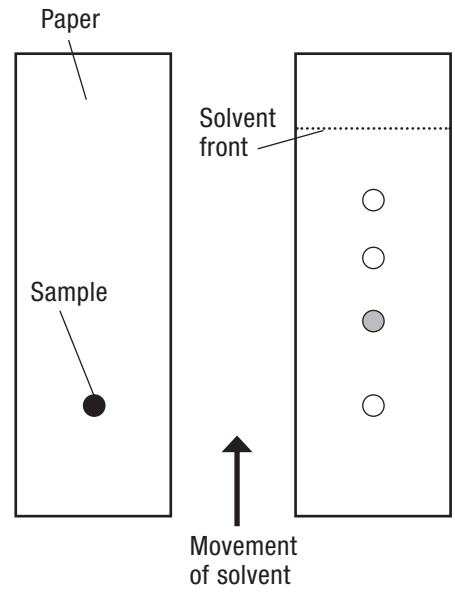

Figure 3

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Components of a mixture can be identified by the distance they travel compared to the distance travelled by the solvent. This relationship is called the **retention factor** and is abbreviated to R_f.

$$R_f = \frac{\text{Distance travelled by component}}{\text{Distance travelled by solvent}}$$

5 How many components were in the sample shown in figure 3?

6 Name three types of chemicals that chromatography can be used to identify.

Thin layer chromatography (TLC) resembles paper chromatography in many ways. It can tell us how many components are in a mixture. TLC is also used to identify compounds in a mixture when the R_f of an unknown compound is compared with the R_f of a known compound.

7 Explain what is meant by R_f.

A TLC plate is a sheet or plate of glass, metal or plastic which is coated with a very thin layer of a solid adsorbent, usually silica or alumina. A spot of the mixture to be analysed is put near the bottom of this sheet. The bottom of the sheet is placed in a solvent with the spot above the level of the solvent. This liquid slowly rises up the sheet by capillary action.

The various components are separated out according to their attraction for the stationary phase and how strongly they are adsorbed. See figure 4.

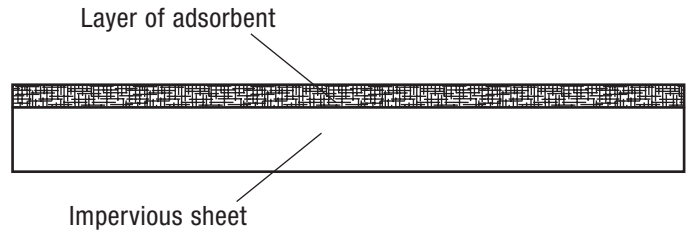

Figure 4

8 Compare ascending paper chromatography with TLC. Ensure you make reference to the difference between *absorption* and *adsorption*.

TRAP

It is easy to confuse adsorption and absorption. ***Adsorption*** is when a foreign substance is attracted to the surface and forms a thin layer with it. ***Absorption*** is when a foreign substance penetrates the body of the absorbing material.

In column chromatography, the column is packed with a particulate adsorbent such as silica gel or alumina. The analyte (mixture being analysed) is placed onto the top of the column. The eluent (solvent used to separate the mixture) is poured in and slowly flows through the packing. See figure 5.

9 Outline how column chromatography works. Include an explanation for why the solvent flows through the column.

10 Decide whether the following statements are true or false.

- **a** All chromatographic techniques involve a stationary phase and a mobile phase.
- **b** Chromatography is only useful for separating and identifying coloured substances.
- **c** Radial chromatography involves a solvent moving upwards.
- **d** *Eluent* is another name for 'solvent'.

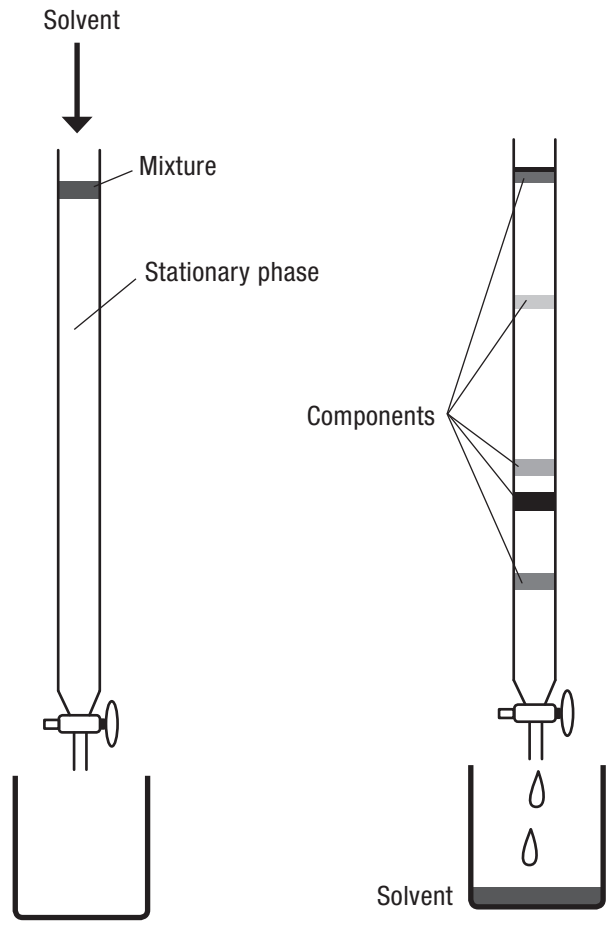

Figure 5

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20 GAS–LIQUID CHROMATOGRAPHY (GLC)

In gas–liquid chromatography (GLC), the stationary phase is a liquid coated onto very small particles of inert solid. This is packed into a narrow tube which is usually referred to as a column, and is normally coiled to save space and made of stainless steel. The mobile phase is an unreactive gas.

1 Name two suitable gases that could be used for the mobile phase of GLC.

A carrier gas (the mobile phase) is allowed to flow through the column until conditions have stabilised. A syringe is used to inject the sample into the machine where it is vapourised using a small oven. The carrier gas transports the analyte (substance being analysed) mixture through the column. The column is contained in a separate oven, with temperatures typically in the range of 50°C to 250°C.

2 Why is the column kept in an oven?

3 Outline why stainless steel is a good material for making a GLC column.

The different components of the analyte will interact differently with the stationary phase in the column. This will cause each compound to have a different retention time, i.e. the time it spends in the column. These retention times are used to identify the substances present. As the components reach a detector, the time of arrival and amount of substance is recorded. Levels as low as in the nanogram range can be detected by more sensitive equipment.

4 What is a nanogram?

THINGS TO KNOW Make sure you know the prefixes for metric units.

GLC cannot always detect and conclusively identify the components in a mixture, so it is often linked to a mass spectrometer. The fragmentation pattern obtained can be compared to others stored in a database.

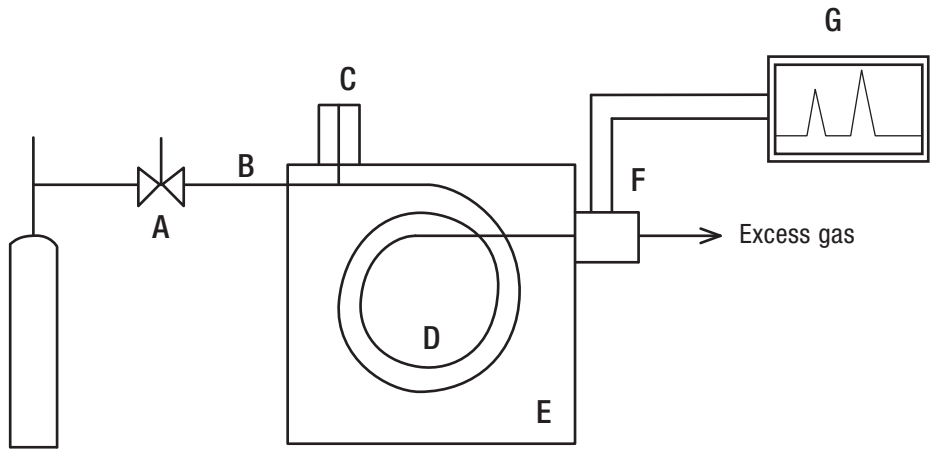

Figure 1

5 Study figure 1 which shows the major components of a gas-liquid chromatogram. Match each part listed below with the correct letter on the diagram.

a carrier gas
b column
c column oven
d detector
e flow controller
f injector port
g recorder

THINGS TO KNOW Make sure you have a clear understanding of important chemical processes such as **chromatography** and relevant key terms such as *stationary phase* and *mobile phase*.

6 Outline the purpose of each of the following parts.

a carrier gas
b column
c column oven
d detector
e flow controller
f injector port
g recorder

7 Discuss the reasons for the column being coiled, rather than straight.

8 What is a database?

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21 ELECTROPHORESIS

Many organic molecules have charges or become electrically charged when dispersed in a fluid. If electrodes are placed in a fluid containing these molecules then the charged particles will move towards the opposite charges.

Electrophoresis is the movement of electrically charged particles through a fluid or gel under the influence of an external electrical field.

TRAP Many students confuse electrophoresis with chromatography. Both use fluids, involve movement of compounds and can separate and identify mixtures of compounds. Electrophoresis uses electricity to attract particles, chromatography doesn't.

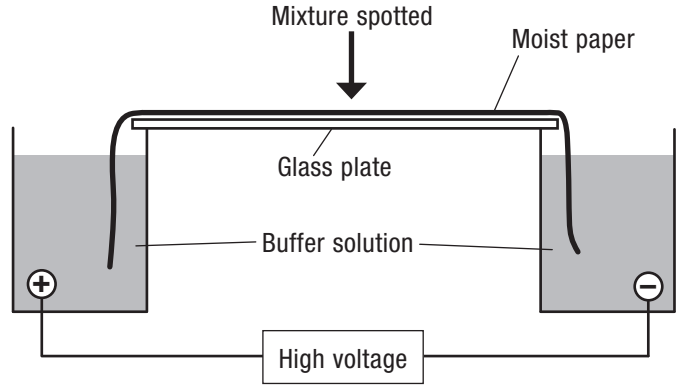

Figure 1

Figure 1 shows the major components used in the technique of paper electrophoresis. The mixture is spotted onto moist paper, the end of which is dipped into buffer solution. A high voltage is applied, causing anions and cations to migrate. This type of electrophoresis is best suited to the analysis of small molecules.

1 Define *anion*.

2 Define *cation*.

TRAP The concept of anions and cations is used in many situations, so it is important that you do not confuse the two types of ion.

3 Why must the paper be moist in paper electrophoresis?

HINT Movement of charge.

Gel electrophoresis uses a gel as the stationary medium. The gel is porous and smaller particles move through it more easily than larger particles do.

Gel electrophoresis can be carried out either horizontally or in columns. It is commonly used to separate and identify DNA mixtures.

Figure 2 shows two views of a typical apparatus used in gel electrophoresis. Holes (sample wells) are cut into the gel and the mixtures of for example DNA fragments are placed in them. Often a dye is added so the progress of the process can be observed. Molecular markers, which are mixtures of DNA with known molecular masses, are also sometimes added. Then a current is applied.

Because the DNA molecules have a negative charge, they are attracted to the positive anode. The greater the charge, the greater the attraction. The DNA then has to be treated chemically before it can be observed.

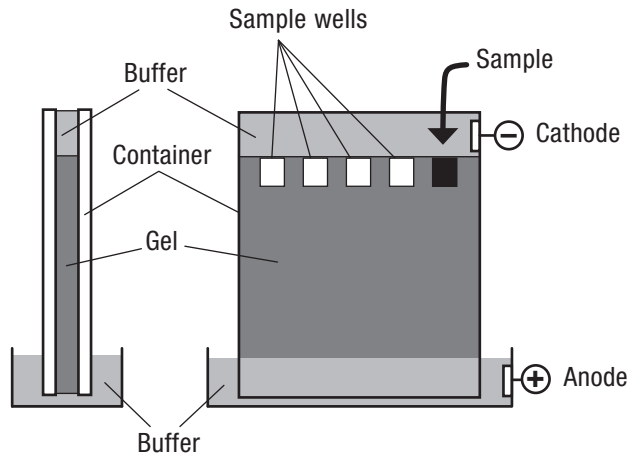

Figure 2

4 What is the role of the buffer solution in the apparatus shown in figure 2?

5 Why do large molecules lag behind when separation starts?

6 Identify a factor other than size that could cause DNA fragments to move at different speeds in electrophoresis.

7 Why are molecular markers used in electrophoresis?

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22 THE MASS SPECTROMETER

The mass spectrometer was invented around 100 years ago. It is a very useful tool for identifying the presence of isotopes of metals and non-metals. It can also be to identify molecular compounds. A mass spectrometer acts as a very accurate balance for finding masses. When molecules are put into a mass spectrometer, they can be broken down into fragments.

Positive ions (cations) travel through a vacuum and are deflected by electric charge and magnetic fields. The degree of deflection can be used to identify the species. The amount of deflection depends on the mass of the particle and its charge.

THINGS TO KNOW

Make sure you have a clear understanding of important chemical processes such as **spectrometry** and know how a spectrometer works.

Here is a sequence followed in a typical mass spectrometer.

Step 1: The (very small) sample is vapourised by heating.

Step 2: The sample is introduced into the ionising chamber.

Step 3: High energy electrons bombard the sample and cause it to form positive ions.

Step 4: The positive ions are accelerated out of the ionising chamber by applying a high voltage.

Step 5: The beam of ions is focused and collimated (turned into a narrow beam).

Step 6: An 'electrostatic analyser' deflects cations of a narrow mass range into the magnetic analyser.

Step 7: The electromagnet is adjusted to select cations of a specific mass to hit the detector.

Step 8: The detector produces a small electric current which is amplified and then recorded.

Step 9: The strength of the electromagnet is adjusted to allow more massive cations to be detected.

Below is a simplified diagram of a mass spectrometer (Figure 1).

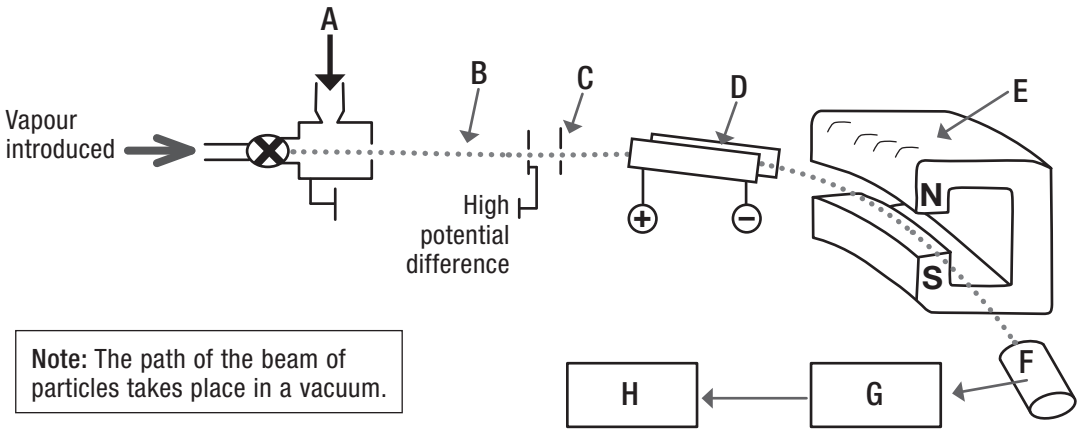

Figure 1

1 Match each part listed below with the correct letter on the diagram.

a acceleration
b amplifier
c beam of electrons
d chart recorder
e collimator
f deflection
g detection
h electromagnet

HINT

Assume that the process is flowing from left to right.

2 Identify the beam of ions on the diagram.

3 What type of charge is on the ions?

4 Why are the particles travelling in a vacuum?

5 Decide whether each statement about mass spectroscopy is true or false.

a It involves destructive testing.
b Only metals can be identified.
c Large quantities of the sample are needed.
d Each compound has a unique mass spectrum.
e It can be used to identify the presence of isotopes of elements.

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23 INTERPRETING MASS SPECTRA

Mass spectroscopy (spectrometry) measures the ratio of the mass to the electrical charge of ions. The mass spectrometer can calculate the relative atomic mass (ram) of atoms, the relative molecular mass (rmm) of covalent compounds and also identify unknown substances.

The detector current (*y*-axis) is proportional to the amount of cations present. The field strength of the electromagnet (*x*-axis) is related to the mass number. Figure 1 illustrates this.

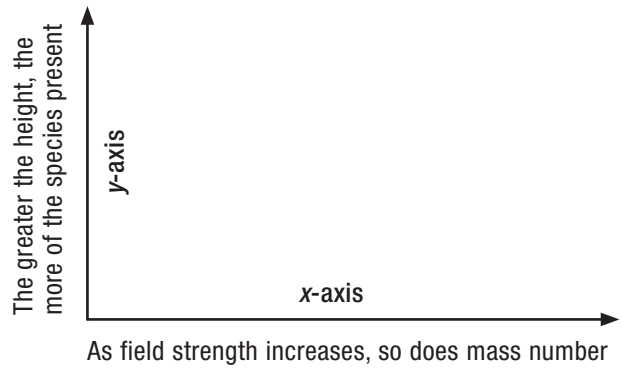

Figure 1

Isotopes are atoms of the same element which differ by the number of neutrons in the nucleus and hence have different mass numbers. The type and quantity of isotopes in a sample of an element can be found using a mass spectrometer. Strictly speaking, the *x*-axis should be labelled the 'mass to charge ratio' – but unipositive ions are selected, so in effect that mass to charge ratio has the same numerical value as the mass.

HINT In a mass spectrum, the *x*-axis shows the mass of the species and the *y*-axis gives the relative amount present.

1 Copy and complete the sentences.

a The _____ the peak on the *y*-axis, the more there is of that cation in the sample.

b As you go from left to right on the *x*-axis, the mass of the cation _____.

2 What is meant by *unipositive*?

WORKED EXAMPLE

The type and quantity of isotopes in a sample of naturally occurring chlorine are recorded. There were two isotopes (see figure 2). ^{35}Cl made up 76% of the sample and ^{37}Cl made up 24% of the sample. What is the relative atomic mass (ram) of the sample?

The formula for calculating ram is shown below.

$$\text{ram} = \frac{(\text{Mass of isotope}_a \times \text{percentage}) + (\text{Mass of isotope}_b \times \text{percentage})}{100}$$

The values are substituted into the equation:

$$\frac{(35 \times 76) + (37 \times 24)}{100} = 35.5 \text{ amu}$$

Therefore the relative atomic mass of the sample is 35.5 amu.

THINGS TO KNOW The relative atomic mass (ram) of an element is the weighted average of the masses of the atoms in a naturally occurring sample of that element.

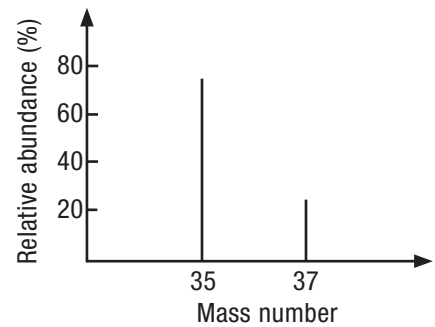

Figure 2

3 A sample of lead was analysed using a mass spectrometer and the results are shown in Table 1. Use these figures to calculate the ram of lead.

Isotope	Atomic mass (amu)	Abundance (%)
^{206}Pb	206	24.6
^{207}Pb	207	22.5
^{208}Pb	208	52.9

Table 1

SLIP It is very easy to make mistakes in calculations of this type. Check your answer carefully and remember that it must be somewhere between the lowest and highest amu figures.

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WORKED EXAMPLE

Figure 3 shows a typical mass spectrum of water (H_2O). Assuming that hydrogen has a ram of 1 amu and that oxygen has a ram of 16 amu, identify the particles represented by the peaks.

Starting with the particle with the largest mass (the parent molecule):

Mass = 18 amu is most likely to be $[H_2O]^+$
(1 + 1 + 16 = 18 amu)

Mass = 17 amu is most likely to be $[HO]^+$
(1 + 16 = 17 amu)

Mass = 16 amu is most likely to be $[O]^+$
(16 amu)

Mass = 1 amu is most likely to be $[H]^+$
(1 amu)

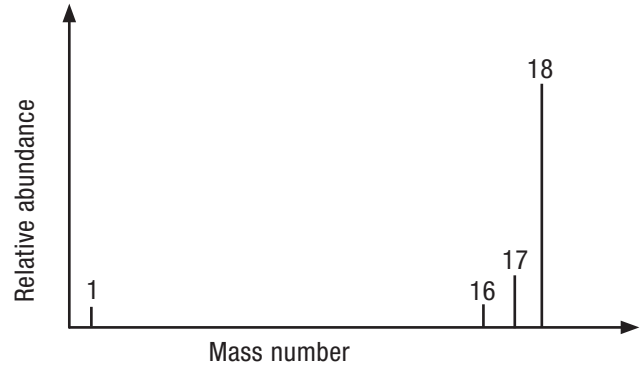

Figure 3

When molecules are processed by a mass spectrometer, they break down into many fragments, each of which will have a different mass-to-charge ratio. Each of these fragments will be shown as a peak on the spectrum. Different molecules will break into different groups and hence each will have a different spectrum. The unique pattern of 'daughter' ions shown in a mass spectrum can be used to work out the identity of the parent molecule. Every covalent molecule has its own individual mass spectrum which can act as a 'fingerprint'.

4 Study figure 4, which shows a typical mass spectrum for a sample of carbon dioxide. Assuming that carbon has a ram of 12 amu and that oxygen has a ram of 16 amu, identify the particles represented by the peaks. Give reasons for your answers.

HINT Study the worked example carefully.

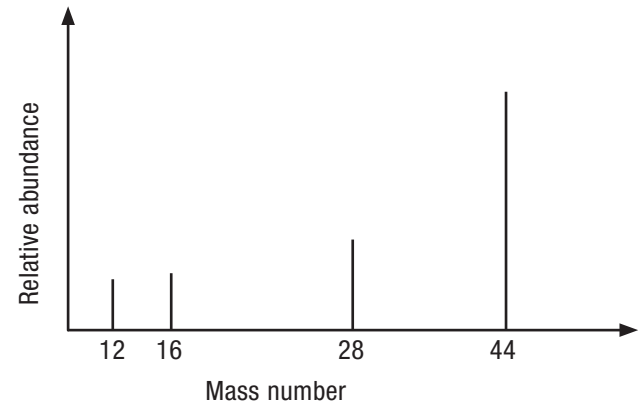

Figure 4

5 Study figure 5 which shows a typical mass spectrum of a sample of hydrogen chloride. Identify the fragments, giving reasons for your answers.

HINT Check the isotopes of chlorine.

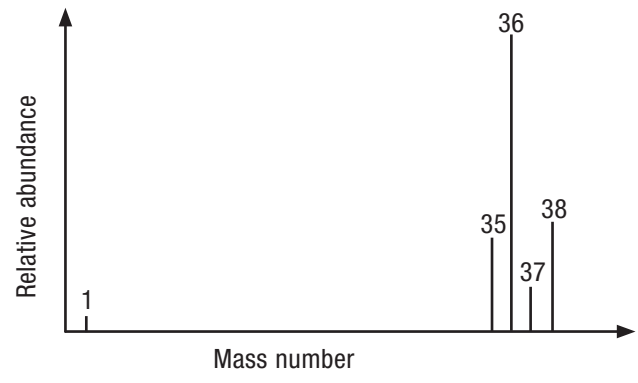

Figure 5

6 Which isotope of chlorine is present in the greater amount in figure 5? Explain your answer.

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24 NUCLEAR CHEMISTRY

Nuclear chemistry deals with the nuclei of atoms. The nucleus contains most of the mass of an atom (protons and neutrons) but occupies very little space. Ernest Rutherford first proposed the existence of the nucleus in 1911. James Chadwick discovered the neutron in 1932. Table 1 shows the properties of the nucleons.

Table 1: Properties of nucleons

	Proton	Neutron
Mass (amu)	1.007 26	1.008 66
Electrical charge	+1	None

1 What does *amu* represent?

The number of protons present in a nucleus of an atom gives the atomic number of that element.

2 Suggest why protons and neutrons are referred to as *nucleons*.

Why does the nucleus hold together? You would expect the positive protons to repel each other. Part of the answer involves 'binding energy'. If you look at the numbers of protons and neutrons in a particular atom and calculate the mass the nucleus should be, you will find that the actual mass of the nucleus is less than the theoretical value. The 'missing' mass (mass deficit) has been transformed into 'binding energy' which holds the nucleons together.

Some elements have isotopes, i.e. their atoms will have the same number of electrons and protons, but different numbers of neutrons. In 1919, Francis Aston used a mass spectrometer to measure mass and percentages of isotopes accurately. Isotopes can be identified by writing the mass number after the name of the element e.g. carbon-14 or uranium-235. The mass quoted in data tables is usually the weighted average of the naturally occurring element.

3 Nucleus A has 15 protons and 16 neutrons, whereas nucleus B has 16 protons and 16 neutrons. Are they isotopes? Explain your answer.

THINGS TO KNOW Make sure you have a clear understanding of the term 'isotope'.

4 Uranium has three naturally occurring isotopes, uranium-234, uranium-235 and uranium-238. A Periodic Table gives the relative atomic mass of uranium as 238.02. What does this tell you about the relative proportions of the isotopes in nature?

TRICK Treat this as an arithmetic question.

Radioisotopes (radioactive isotopes) are isotopes of an element that have unstable nuclei and emit nuclear radiation to become more stable. There are three major types of nuclear radiation: alpha (α), beta (β) and gamma (γ). Strictly speaking, alpha and beta are particles and only gamma is truly radiation.

- **Alpha** particles consist of helium nuclei, i.e. particles containing two protons and two neutrons. These travel at about 5–10% of the speed of light and only travel a few centimetres in air. They can be stopped by a thin paper sheet.
- **Beta** particles are high energy electrons that can travel almost at the speed of light and can be stopped by a few millimetres of aluminium.
- **Gamma** rays are a form of high energy electromagnetic radiation (emr) and can only be stopped by a considerable thickness of concrete or lead.

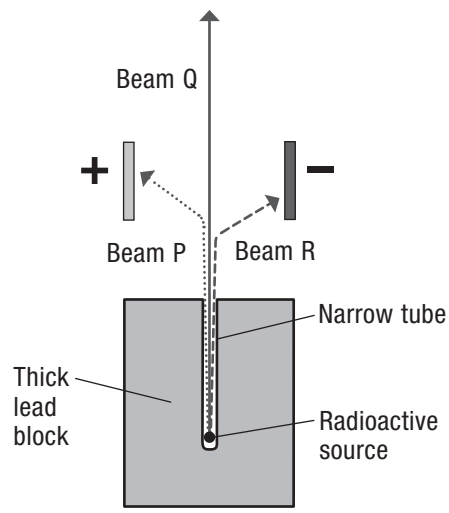

Figure 1

5 Draw up a table with the column headings 'Name', 'Nature of radiation', 'Electrical charge', 'Speed' and 'Penetrating power'. Use it to summarise the features of the three major type of nuclear radiation.

6 The nucleus makes up 99.97% of the mass of an atom but only occupies approximately 0.001% of its volume. True or false?

7 A radioactive source which emits alpha, beta and gamma radiation was placed in a lead block as shown in figure 1.

a Identify radiation P. **b** Identify radiation Q. **c** Identify radiation R.

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When an alpha particle is ejected from the nucleus, the atomic number decreases by two and the mass number decreases by four. Loss of a beta particle causes no change in mass number but the atomic number increases by one. This sounds strange, but it is a result of a neutron splitting into a proton and an electron. Emission of gamma radiation results in the nucleus having less energy.

In equations, an alpha particle is shown as $^{4}_{2}\text{He}$ and a beta particle as $^{0}_{-1}\text{e}$.

WORKED EXAMPLES

Example 1: Uranium-238 can emit an alpha particle to form thorium-234. Write a balanced equation for this nuclear reaction.

The first step is to show all species present with symbol, atomic number and mass number.

Uranium-238 is represented by $^{238}_{92}\text{U}$, alpha particle is $^{4}_{2}\text{He}$, and thorium-234 is $^{234}_{90}\text{Th}$.

Next write down the equation: $^{238}_{92}\text{U} \rightarrow {}^{4}_{2}\text{He} + {}^{234}_{90}\text{Th}$

Now check that the numbers balance:

- Reactants – Atomic number = 92, Mass number = 238
- Products – Atomic number = 90 + 2 = 92, Mass number = 234 + 4 = 238

Yes, the numbers balance so the equation is: $^{238}_{92}\text{U} \rightarrow {}^{4}_{2}\text{He} + {}^{234}_{90}\text{Th}$

Example 2: Protactinium-234 can emit a beta particle to form uranium-234. Write a balanced equation for this nuclear reaction.

Protactinium-234 is represented by $^{234}_{91}\text{Pa}$, a beta particle is $^{0}_{-1}\text{e}$, and uranium-234 is $^{234}_{92}\text{U}$.

$^{234}_{91}\text{Pa} \rightarrow {}^{0}_{-1}\text{e} + {}^{234}_{92}\text{U}$

- Reactants – Atomic number = 91, Mass number = 234
- Products – Atomic number = 92 – 1 = 91, Mass number = 234

The equation is $^{234}_{91}\text{Pa} \rightarrow {}^{0}_{-1}\text{e} + {}^{234}_{92}\text{U}$

8 Write balanced equations for the following nuclear reactions.

a Radium-226 decaying to radon-222 by means of emission of an alpha particle

b Carbon-14 emits a beta particle to become nitrogen-14

c Phosphorus-32 decaying to sulfur-32

TRAP When writing equations for nuclear decay, you must be careful that all the atomic and mass numbers balance and that you use the correct symbols and numbers for species that are present.

Sometimes nuclei are so unstable that rather than just releasing small fragments, they split into smaller nuclei to form new atoms, simultaneously releasing energy. This process is called **fission**. The small decrease in mass that occurs releases huge amounts of energy in accordance with Einstein's famous equation, $E = mc^2$**.** Fission is the splitting of nuclei, releasing energy.

9 Rewrite the equation $E = mc^2$ in words.

If uranium-235 – which is much rarer than uranium-238 – is bombarded by neutrons, it can split up violently releasing energy and more neutrons as shown in the following equation.

$$^{0}_{1}\text{n} + {}^{235}_{92}\text{U} \rightarrow [{}^{236}_{92}\text{U}] \rightarrow {}^{90}_{38}\text{Sr} + {}^{143}_{54}\text{Xe} + 3{}^{0}_{1}\text{n}$$

10 Rewrite the equation above in words.

In the reaction above showing the bombardment of uranium-235 by neutrons, three high energy neutrons ($^{1}_{0}\text{n}$) are released. In turn, these three neutrons hit another three nuclei, causing each one to split, and so on. If this 'chain reaction' is not controlled, it can cause a nuclear explosion. Nuclear reactors moderate this reaction and use the heat generated to produce steam to drive turbines to generate electricity. Radioisotopes can also be made in nuclear reactors.

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25 RADIOACTIVITY AND HALF-LIFE

For any radioisotope, the amount of radiation it gives off will gradually decrease. This decrease in activity is described using the concept of 'half-life'. Each radioactive species has its own characteristic half-life. There are two ways to describe half-life.

Half-life is the time required for:

- half the nuclei in a sample of a radioisotope to undergo radioactive decay.
- the activity of a sample of a radioisotope to decrease to one half of its original value.

Figure 1 illustrates how the amount of a radioisotope varies over time.

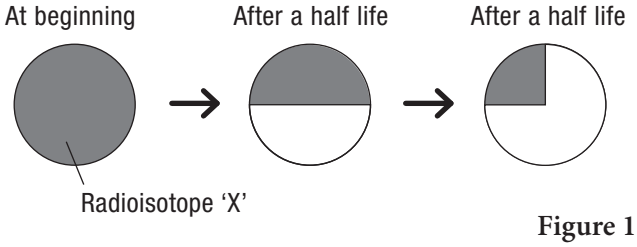

Figure 1

THINGS TO KNOW

Half-life can be confusing. Use several sources to gain a better understanding of this concept.

1 Copy and complete the following passage using figure 1 and these words:
disappeared, eighth, half, original, quarter, radioisotope, time.

Half-life is the _____ taken for the concentration of a _____ in a sample to decrease to half of its _____ value. The rest of the sample hasn't _____, but has changed to another substance. After another half-life a _____ will be left and after another half-life there will be an _____ left, and so on.

2 Iodine-131 has a half-life of 8.1 days. Calculate the proportion of iodine-131 remaining in a sample after 40.5 days.

HINT

One way to approach this is: 'After 8.1 days there will be half, after 16.2 days there will be a quarter...'

The half-life can also be defined as the time it takes for the amount of radioactivity being given off to fall to half of the initial value. Typical questions will ask you to draw a graph of time versus radioactivity and/or interpret data from a graph.

3 The table at the right shows a typical set of readings of the radioisotope sodium-24. Use the data to draw a scatter graph. Then draw a line of best fit.

Time (hours)	Counts per minute
0	400
1	260
2	160
3	100
4	65
5	42
6	25
7	10
8	8

HINT

Plot the independent variable (time) on the *x*-axis and the dependent variable (counts) on the *y*-axis. Label the axes and include units.

4 Refer to your graph to answer the following questions.

a What is the half-life of sodium-24?
Show on your graph how you obtained this figure.

b What percentage of the original isotope remains after 4½ hours?
Explain your answer.

5 Are the following statements true or false?

a Energy changes in nuclear reactions are usually much less than those in chemical reactions.

b The rate of decay of radioisotopes cannot be changed by physical or chemical means.

c All half-lives of radioisotopes are approximately the same.

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26 RISK ASSESSMENTS

Practical work plays a major part in all Chemistry courses. All practical activities have potential hazards and a risk assessment should be carried out before proceeding with any practical work. Your teacher will have completed a risk assessment before you undertake any practical activity. However, you are expected to have an awareness of health and safety issues relevant to the Chemistry laboratory.

You should be able to complete the following for practical activities:

- identify possible hazards
- assess the risks that might arise from these hazards
- describe appropriate measures to minimise the risks.

THINGS TO KNOW

Health and safety are not simply theoretical concepts. Think about the associated issues every time you plan an experiment. Practise the skills of hazard identification and risk minimisation every time you work in a laboratory.

Possible hazards include:

- glassware – can be hot or break when dropped
- chemicals – can be toxic, corrosive, carcinogenic or cause allergic response
- liquids – can be spilt or 'spit' from heated mixtures
- solvents – flammable, toxic, narcotic
- naked flames e.g. Bunsen burner
- dust or fumes – can be generated during burning
- water – can be spilt on the floor, making it slippery.

1 Match each term with the correct meaning.

Term	Meaning
a allergy	• Danger
b carcinogenic	• Chemical that dulls the senses
c corrosive	• Poisonous
d flammable	• Make as small as possible
e narcotic	• Damages materials on contact
f toxic	• Exaggerated response by the body to foreign substance
g hazard	• Easily set on fire
h minimise	• Causes cancer

2 For each hazard listed below, outline how you could minimise the risks.

a Glassware

b Chemicals

c Heating liquids

d Solvents

e Naked flames

f Dust/fumes

3 Think of a Chemistry practical you have carried out. Briefly describe it and identify the possible hazards and the steps you took to ensure that you (and others) were at minimal risk.

4 A sign outside a Science Laboratory stated: 'No school bags allowed inside'. Give a reason for this rule.

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27 PRACTICAL WORK

As well as carrying out practical work safely, you need to record your work. The format required may vary slightly, but in general, each practical should have a title, an aim, an equipment list, a diagram, a method (procedure), results and a conclusion.

1 Match each term with the correct meaning.

Term	Meaning
a Aim	• Scientific drawing of apparatus
b Conclusion	• Name of the experiment
c Diagram	• How you carried out the practical work
d Equipment	• What happened
e Method	• Summary of results
f Results	• What you are trying to find out
g Title	• Apparatus that was used

THINGS TO KNOW

Learn the key parts of a practical report and make sure you understand the purpose of each part. This wil help you stay focused during the practical work as well as contribute to the clarity of your report.

2 Outline any other aspects that could be added to this format.

The **title** should be as short as possible and refer to the work carried out. It should help you remember what you did when you come back to revise it.

3 State one other thing that should accompany the title.

The **aim** is a statement saying what you are trying to find out in the experiment.

4 Give another description for *aim*.

The **equipment list** is often optional and can be omitted if the diagram shows all apparatus used. The **diagram** must be clearly labelled, showing what the equipment is and how it is used.

5 List two more considerations when drawing a diagram.

The **method** describes how the experiment was performed. As part of it, you should identify possible dangers involved in the practical and how you plan to minimise them. It should be written in the past, impersonal tense.

6 Explain what is meant by the *past, impersonal tense*. Include an example.

HINT

'The mixture was heated.'

The **results** should include your observations of what happened when you carried out the experiment. Note that observations are not only what you see, they can include a sound or a smell.

7 Describe three different ways that results can be recorded.

In the **conclusion**, you summarise all you have learned from the experiment. Note any problems or possible improvements to the practical. Ask yourself, *What could be done to make the outcomes from the practical better?* If in doubt, 'Do it again and get averages' is a good default approach. You need to identify any patterns or trends in your results. In your conclusion, you may also be required to discuss how you ensured your method was a fair test (i.e. valid) and did indeed test the aim.

8 What is meant by *default approach*?

9 Think of a Chemistry practical you have carried out. Briefly describe it, identifying the variables and the steps you took to make certain that the procedure was a fair test.

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28 VARIABLES AND CONTROLS

When experiments are being carried out, we concentrate on observing and recording what happens when conditions change. Factors that can be changed during an experiment are called **variables**. A variable is any factor that can exist in differing amounts or types during an investigation. Usually only one variable is changed at a time. There are three types sorts of variables: **independent**, **dependent** and **controlled.**

- The **independent variable** is the major part of the hypothesis (belief) that you are testing. It is the condition that you change to see what happens.
- The **dependent variable** is the thing that changes when the independent variable is changed.
- A **control (controlled variable)** is one that is kept the same in each experiment.

THINGS TO KNOW

Hypothesis: an explanation for an observation that can be tested by further investigation.

	Variables		
Hypothesis: Different types of capsules and tablets dissolve at different rates	**Independent:** Type of capsules or tablet	**Dependent:** Time for reaction to occur	**Controlled:** Volume of water, Temperature

1 State two other possible variables in the investigation outlined in the table above.

RATES OF REACTION

Marble chips are mainly calcium carbonate ($CaCO_3$). Carbonates react with acids to give off the gas carbon dioxide (CO_2):

acid + carbonate → salt + water + carbon dioxide

A student was given the task of investigating whether or not the size of marble chips alters their rate of reaction with an acid. The student took 10 g of marble chips and divided them into two equal portions. One half was ground into smaller particles and the other half left as large lumps. The student added the samples of marble chips to dilute hydrochloric acid as shown in figure 1. He timed how long it took for the reaction to take place. The following table shows his results.

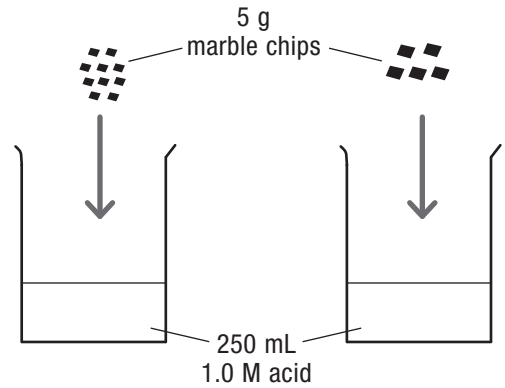

Figure 1

Size of marble chips	Large	Small
Time (sec)	632	421

2 How would the student know when the reaction was complete?

3 What is the independent variable in this investigation?

4 What is the dependent variable in this investigation?

5 What are the controlled variables in this investigation?

6 Give a hypothesis for this investigation.

7 Write a word equation for the reaction in this investigation.

8 Write a balanced formula equation for this reaction.

TRAP

Make sure you know and understand the definitions for the terms used in questions 3, 4, 5 and 6.

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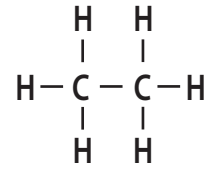
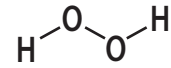
29 FORMULAE OF COMPOUNDS

There are three major types of chemical formulae; molecular, empirical and structural. Molecular formulae only apply to covalent compounds and tell the actual numbers of atoms of the elements present. Empirical formulae express the relative numbers (i.e. a ratio) of atoms present. (The molecular formulae and the empirical formula of a compound can be the same.) The formulae of ionic compounds are always empirical. Structural formulae show the types and number of atoms present and also how these atoms are bonded and arranged in a molecule.

Name	Molecular formula	Empirical formula	Structural formula
Hydrogen chloride	HCl	HCl	H—Cl
Ethane	C_2H_6	CH_3	
Hydrogen peroxide	H_2O_2	HO	

1 Give the names and numbers of atoms of each element present in:

a hydrogen chloride.

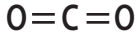
b ethane.

c hydrogen peroxide.

HINT Look at structural formula to get the molecular formula. Then think of the 'lowest common denominator' when changing molecular formula to empirical formula.

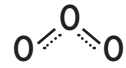
2 Using the table above as a guide, draw up a similar table for each substance shown, completing the molecular formular and the empirical formula.

Carbon dioxide

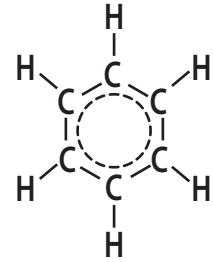


Oxygen

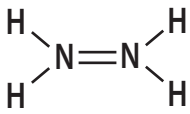
Ozone



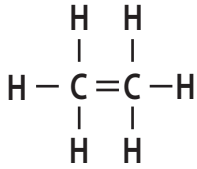
Benzene



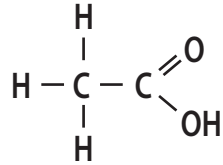
Hydrazine



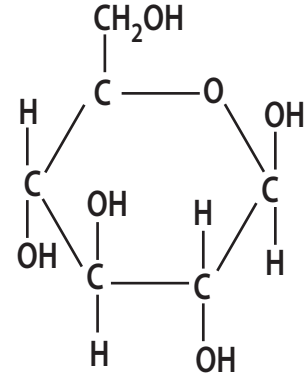
Ethene



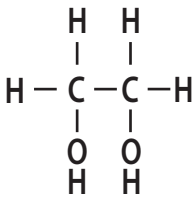
Ethanoic acid



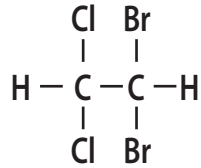
Glucose



Ethan-1, 2 diol



1, 1-dibromo, 2, 2-dichloroethane



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30 CALCULATING FORMULAE

Ionic compounds and macromolecules (e.g. covalent network structures) do not have molecular formulae**.** Instead, their empirical formula is used. This doesn't give the actual number of atoms present, which could be a very large number, but the simple whole number ratio of atoms – the empirical formula of a compound states the relative number of each type of atom present.

When carrying out analysis of the composition of a chemical compound, only the empirical formula can be found. A typical quantitative investigation into a compound will reveal either the masses or the relative percentages of individual elements present. The procedure for working out the empirical formula is the same in both cases. The number of moles of each element present is calculated and then the ratio.

WORKED EXAMPLE

Calculating empirical formula

A sample of an oxide of carbon was found to contain 0.544 g of carbon and 0.726 g of oxygen. What is the empirical formula of this compound? (Atomic mass: carbon = 12.0 and oxygen = 16.0)

• Write down the symbols of the elements present:	C		O
• Write down the amounts of the elements present:	0.544 g		0.726 g
• Divide by the molar mass: *(Gives number of moles)*	÷ 12.0 g = 0.453		÷ 16.0 g = 0.454
• Divide the larger number by the smaller number: *(Gives ratio)*		$\frac{0.454}{0.453} = \frac{1}{1}$ ratio is 1 : 1	

Hence the formula is CO.

1 Work out the empirical formulae for the following samples of compounds.

a 7.99 g of carbon and 2.01 g of hydrogen. (Atomic mass of hydrogen = 1.00)

b 1.88 g of carbon, 0.625 g of hydrogen and 2.50 g of oxygen

2 5.46 g of phosphorus and 7.04 g of oxygen (Atomic mass of phosphorus = 31.0)

THINGS TO KNOW

Ensure that you know how to find out the atomic mass and molar mass of an element from a Periodic Table or other source.

If you know the empirical formula and molecular mass of a compound, you can work out the molecular formula.

WORKED EXAMPLE

A hydrocarbon has the empirical formula CH_3 and a molecular mass of 30.0 amu. What is its molecular formula?

• Work out the mass according to the empirical formula: *(One carbon and three hydrogens)*	$(1 \times 12.0) + (3 \times 1.00) = 15.0$ amu
• Divide the molecular mass by the formula mass: *(This gives the number of 'units' in the molecular formula)*	30.0 amu ÷ 15.0 amu = 2
• Multiply the empirical formula by this number.	$(CH_3) \times 2 = C_2H_6$

Hence there are two CH_3 'units' in the molecule, so the molecular formula is C_2H_6.

3 Work out the empirical formulae and then the molecular formulae for the following samples of compounds.

a A hydrocarbon with molar mass of 78.9 g; carbon present was 1.29 g and hydrogen was 0.108 g

b Compound containing carbon (0.84 g), hydrogen (0.14 g) and oxygen (1.12 g) with a molar mass of 180.0 g

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A SELECTED ANSWERS

1 The atmosphere

1 [in order from top] N_2, oxygen, Ar, carbon dioxide, Ne
2 The area on Earth where life exists
3 **a** Atmosphere **b** Soil/ground
4 Ammonium NH_4^+; Nitrate NO_3^-; Nitrite NO_2^-
5 **a** Change ammonium and nitrite compounds to nitrates **b** Change nitrogen in soil into ammonium compounds **c** Break down plants and animals forming ammonium compounds **d** Remove nitrates in soil to atmosphere as nitrogen
6 Animals eat plants, excretion from animals 'feeds' plants
7 The layer of air that surrounds the Earth

2 Oxygen

1 Gas is pushed through the delivery tube, displaces water and fills up the gas jar
2 $2HgO_{(s)} \rightarrow 2Hg_{(l)} + O_{2(g)}$
3 Mercury and mercury compounds are toxic substances
4 $H_2O_{2(aq)} \rightarrow H_2O_{(l)} + \frac{1}{2}O_{2(g)}$
5 A substance that speeds up a reaction but is not used up in the process
6 More strongly or quicker. Air is only 20% oxygen
7 $S_{(s)} + O_{2(g)} \rightarrow SO_{2(g)}$ **8** That it is acidic **9** **a** 96 L **b** 3.77 g
10 Lowers the levels of harmful UV rays reaching the Earth's surface, reduces rates of skin cancer, CFCs react with ozone

3 Carbon dioxide

1 Poisonous **2** $CO_{2(g)} + Ca(OH)_{2(l)} \rightarrow CaCO_{3(s)} + H_2O_{(l)}$
3 Photosynthesis is left to right. Respiration is right to left
4 It will 'undissolve' and enter the atmosphere
5 **b** Material that is or was living **6** [in order] D, C, A, B
7 [in order] phytoplankton, biological, unicellular, light, metabolism, chlorophyll, oxygen, live, ocean, sink

4 Atmospheric pollution

1 Something with a smell that annoys
2 Small particles suspended in air – dust **3** $313\ 170 \times 10^6$ L
4 $212\ 795 \times 10^6$ g **5** Carbon dioxide
6 Photosynthesis, dissolving in water
7 Coal, oil, natural gas **8** CFCs (5 490)
9 Pushes the liquid contents of a can out in the form of a spray of droplets
10 Oxygen **11** Chlorine, fluorine and carbon
12 Energy causes O_2 molecules to split into activated single atoms. These combine with other O_2 molecules to form ozone (O_3) molecules
13 Screens, stratospheric, life, hole, increase
14 $SO_{2(g)} + H_2O_{(l)} \rightarrow H_2SO_{3(aq)}$ **15** HNO_3
16 Organisms that live in water
17 Low boiling point, changes from liquid to gas easily
18 Trouble with breathing, e.g. lung diseases **19** Smoke + fog
20 The Sun **21** Oxides of nitrogen

5 Gases

1 **a** collision **b** elastic **c** minute **d** negligible **e** discrete **f** translation **g** kinetic **h** chaotic
2 As volume increases, pressure decreases. as pressure increases, volume decreases
3 A fixed mass of gas is sealed in a graduated tube. Pressure is varied and the volume at different pressures is measured
4 fill, decreases, pressure, volume, temperature, collide, halved, increased
5 $P_1V_1 = P_2V_2$; $150 \times 200 = 275 \times V_2$
$V_2 = (150 \times 200)/275 = 109$ mL
6 Mass and pressure
7 As temperature increases, volume increases
8 Vary the temperature of liquid and record the volume of gas at different temperatures
9 volume, raised, kinetic, rapidly, collide, constant, fixed, relationship
10 For a fixed mass of gas at constant pressure, volume divided by temperature is a constant. Or, volume is proportional to temperature.
11 $(V_1/T_1) = (V_2/T_2)$ $18.5/(273 + 15) = V_2/(273 + 30)$
$18.5/288 = V_2/303$ $V_2 = (303 \times 18.5)/288 = 19.5$ L
12 220 K = -53°C **13** The particles get slower

6 Gas Laws

1 For a fixed mass of gas, the product of pressure and volume divided by temperature is a constant
2 $(P_1V_1/T_1) = (P_2V_2/T_2)$ $(199 \times 0.46/313) = (101.3 \times V_2/273)$
$V_2 = 0.788$ L
3 55.84 kPa **4** $n = PV/RT = (1200 \times 1.25)/(8.314 \times 298) = 0.61$
5 0.011 moles = 0.46 g, molar mass = 41.8 g

7 Molar volume of gases

1 No fixed volume, no fixed shape, made up of discrete particles
3 **a** $n = 3.7/22.4 = 0.165$ **b** $49.5/22.4 = 2.21$ **c** $0.25/22.4 = 0.011$
4 **a** $8.7 \times 24.5 = 213.15$ L **b** $0.94 \times 24.5 = 23.0$ L **c** $2.5 \times 24.5 = 61.25$ L
5 **a** $8.51/24.5 = 0.347$ mol = 32.9 g **b** 76.8 g **c** 40.7 g
6 This logic doesn't work with the Celsius scale. If you convert to Kelvins, 20°C to 40°C becomes 293 K to 313 K.

8 Catalysts

1 Valence or bonding
2 Very low, energy present at room temperature is enough
3 e.g. striking a match, lighting a Bunsen burner, lighting magnesium
4 Should be the reverse of figure 1, i.e. products at a higher energy than reactants
5 Catalysed is dotted (lower) line and uncatalysed is solid (upper) line
6 $H_2O_{2(l)} \rightarrow H_2O_{(l)} + \frac{1}{2}O_{2(g)}$ **7** Manganese dioxide
8 Can be produced at a reasonable cost – profitable.
9 Ethene (ethylene)
10 So combustion or other reactions with oxygen don't take place
11 Lower temperature required, speeds up reaction and hence lower costs
12 $N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$ **13** NH_4^+ **14** HNO_3
15 $2NH_{3(g)} + 2\frac{1}{2}O_{2(g)} \rightarrow 2NO_{(g)} + 3H_2O_{(l)}$
16 $NO_{(g)} + \frac{1}{2}O_{2(g)} \rightarrow NO_{2(g)}$
17 $2NO_{2(g)} + O_{2(g)} + H_2O_{(l)} \rightarrow 2HNO_{3(aq)}$
18 $S(s) + O_{2(g)} \rightarrow SO_{2(g)}$ **19** $SO_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow SO_{3(g)}$
20 V_2O_5 **21** $SO_{3(g)} + H_2O_{(l)} \rightarrow H_2SO_{4(g)}$ **22** **a** True **b** True

9 Water

1 Liquid, transparent, colourless
2 $1000 \times 1000 \times 1000 \times 1 \times 4.18 = 4\ 180\ 000\ 000$ J = 4 180 MJ
3 Perspiration
4 Convection, oxygen, surface, dense, displaced, cycle
5 Not usual, uncharacteristic **6** 1 $g.mL^{-1}$
7 As water reaches its freezing point, it expands
8 **a** False, true figure is around 70% **b** True

10 Structure and bonding of water

1 Covalent
2 Should show the δ^+ ends of water surrounding a negative charge.
3 **a** $NaOH_{(s)} + H_2O_{(l)} \rightarrow Na^+_{(aq)} + OH^-_{(aq)}$

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b $KOH_{(s)} + H_2O_{(l)} \rightarrow K^+_{(aq)} + OH^-_{(aq)}$
c $Na_2SO_{4(s)} + H_2O_{(l)} \rightarrow 2Na^+_{(aq)} + SO_4{}^{2-}_{(aq)}$
d $NH_4Cl_{(s)} + H_2O_{(l)} \rightarrow NH_4{}^+_{(aq)} + Cl^-_{(aq)}$
e $K_3PO_{4(s)} + H_2O_{(l)} \rightarrow 3K^+_{(aq)} + PO_4{}^{3-}_{(aq)}$

4 polar, electrical, dissolved, ions, dissociation, hydrated, conductivity
5 Only ionises partially **7** 0.65 $mol.L^{-1}$
8 Greater, as more ions are present.
$MgCl_{2(s)} + H_2O_{(l)} \rightarrow Mg^{2+}_{(aq)} + 2Cl^-_{(aq)}$
9 hydrochloric acid HCl, nitric acid HNO_3, sulfuric acid H_2SO_4
10 water H_2O, hydrofluoric acid HF, ethanoic acid CH_3COOH

11 Water quality

1 Fish, plants
2 Substance (usually harmful) that shouldn't be there
3 Collect water on roof and store in a tank
4 Area in which rain falls and is collected
5 purifying, sedimentation, fine, aluminium, filter, disease, chlorine, carbon
6 b, c, e, d, a **7** Generates salt/salty water, needs large scale plant

12 Water pollution

1 Production of wood pulp and paper; bleaches such as chlorine can kill organisms
2 Diesel, oil, harmful chemicals can find their way into the water table; contamination of water supplies
3 Decrease oxygen content, harm aquatic life, cause algal bloom
4 Particulate matter (dust) can settle in bodies of water, possible leakage, e.g. of radioactive materials; oxides of nitrogen formed can contribute to acid rain
5 Fish die and unwanted organisms can flourish

13 Solutions

1 Heat, stir, have solid in small pieces
2 Any quantities can mix together
3 Evaporate water leaving salt behind
4 Distil it, collect water **5 a** C_2H_6O **b** C_6H_6
6 polar, best, solute, non-polar **7 a** Endothermic **b** Exothermic
8 Having very strong 3-dimensional bonds **9** Sodium chloride
10 a ionic **b** covalent network **c** covalent molecular **d** polar covalent
11 Should show Na^+ forming bond with the δ^- part of water.
12 Soluble: $NaNO_3$, KCl, $MgSO_4$, NH_4Cl, Li_2CO_3, $AgNO_3$, BaI_2, $MgBr_2$, $Ca(NO_3)_2$, $(NH_4)_2CO_3$, NH_4Br, $Pb(NO_3)_2$, K_2CO_3,
Insoluble: $PbCl_2$, CaF_2, $HgSO_4$, $Ca_3(PO_4)_2$, $Li_3(PO_4)$

14 Saturated solutions

1 Saturated, solute, temperature, unsaturated **3** Increases rapidly
4 Increases very slightly **5 a** 116 g **b** 40 g **c** 39 g **d** 37 g
6 Potassium nitrate would come out of solution: 117 – 39 = 88 g
7 Decreases as temperature rises **8** Increases as temperature rises
9 Both oxygen and carbon dioxide are less soluble in higher temperatures and hence fish and plant life are affected adversely
10 Increase in pressure increases solubility of gas
11 Pressure is reduced; carbon dioxide gas comes out of solution, fizz results
12 At depth, the pressure increases and nitrogen dissolves in the bloodstream. As a diver rises to the surface, nitrogen comes out of the bloodstream. If the diver rises too quickly, the gas can bubble out causing damage to the body
13 More oxygen gets into the bloodstream very quickly, metabolism speeds up
14 Gold is too expensive and too soft **15** 18 ÷ 24 = 75%
16 Low melting point, good electrical conductivity
17 A glass is an inorganic product of fusion which has been cooled to the solid state without crystallising
18 a wine **b** soda water **c** air **d** stainless steel

15 Solutions, concentration and strength

1 Divide relative numbers of solute particles by volume.
A = 11/500 = 0.022 B = 6/300 = 0.02 C = 7/400 = 0.017 5
D = 4/250 = 0.016 Hence D, C, B and then A
2 a 5/0.5 = 10 $g.L^{-1}$ **b** 38 $g.L^{-1}$ **c** 20.65 $g.L^{-1}$ **d** 25.6 $g.L^{-1}$
3 a 2.5 $mol.L^{-1}$ **b** 0.5 $mol.L^{-1}$ **c** 0.87 $mol.L^{-1}$ **d** 0.4 $mol.L^{-1}$
4 $NaOH_{(s)} + H_2O_{(l)} \rightarrow Na^+_{(aq)} + OH^-_{(aq)}$
5 When a molecule has areas of slightly different electrical charge.
6 Reactants are present in greater numbers than products.
7 X is strong, all molecules break up. Y is weak as only small number break up.
8 a $HCl_{(aq)} + H_2O_{(l)} \rightleftharpoons H_3O^+_{(aq)} + Cl^-_{(aq)}$
b $HNO_{3(l)} + H_2O_{(l)} \rightleftharpoons H_3O^+_{(aq)} + NO_3{}^-_{(aq)}$
c $H_2SO_{4(l)} + 2H_2O_{(l)} \rightleftharpoons 2H_3O^+_{(aq)} + SO_4{}^{2-}_{(aq)}$
9 a $CHOOH_{(l)} + H_2O_{(l)} \rightleftharpoons CHOO^-_{(aq)} + H_3O^+_{(aq)}$
b $CH_3COOH_{(l)} + H_2O_{(l)} \rightleftharpoons CH_3COO^-_{(aq)} + H_3O^+_{(aq)}$
c $H_2CO_{3(aq)} + H_2O_{(l)} \rightleftharpoons HCO_3{}^-_{(aq)} + H_3O^+_{(aq)}$
d $H_3PO_{4(l)} + 3H_2O_{(l)} \rightleftharpoons PO_4{}^{3-}_{(aq)} + 3H_3O^+_{(aq)}$
10 a $NH_4OH_{(s)} + H_2O_{(l)} \rightleftharpoons NH_4{}^+_{(aq)} + OH^-_{(aq)}$
b $KOH_{(s)} + H_2O_{(l)} \rightleftharpoons K^+_{(aq)} + OH^-_{(aq)}$
c $Ca(OH)_{2(s)} + H_2O_{(l)} \rightleftharpoons Ca^{2+}_{(aq)} + 2OH^-_{(aq)}$

16 The Earth's Resources

1 a Layer of gas surrounding the Earth **b** Zone where life exists **c** Will eventually run out **d** Part of the Earth where water is present **e** Solid outer layer of the Earth **f** Made by humans
3 Gold, silver, tin, platinum, copper
4 a Oxygen, O **b** Carbon, C and oxygen, O **c** Chlorine, Cl **d** Silicon, Si and oxygen, O **e** Sulfur, S

17 Extraction of metals

1 Occurring, useful, deposit, reserve, impurities.
2 $PbS_{(s)} + O_{2(g)} \rightarrow Pb_{(s)} + SO_{2(g)}$
3 $Cu_2S_{(s)} + O_{2(g)} \rightarrow Cu_{(s)} + SO_{2(g)}$
4 $SnO_{2(s)} + C_{(s)} \rightarrow Sn_{(s)} + CO_{2(g)}$
5 $3CO_{(g)} + Fe_2O_{3(s)} \rightarrow 2Fe_{(s)} + 3CO_{2(g)}$
6 $3O^{2-} \rightarrow 1½O_{2(g)} + 6e^-$ **7** $2Al^{3+} + 6e^- \rightarrow 2Al_{(s)}$
8 a $Cl^- \rightarrow ½Cl_{2(g)} + e^-$ **b** $Na^+ + e^- \rightarrow Na_{(s)}$

18 Transition metals

1 Two in the first shell, then maximum of eight in next two shells
2 Energy required to remove an electron from an atom to form a positive ion, usually quoted per mole
3 Shiny, ductile, malleable, good electrical and thermal conductors
4 React with non-metals, form basic oxides, form positive ions, react with acids
5 Group of elements from 57 to 71, also known as the 'rare earths'
6 Positive ion
7 Form coloured compounds, have more than one oxidation state
8 a $KMnO_4$ **b** Cr_2O_3 **c** $Fe(OH)_2$ **d** $Co(NO_3)_2$
9 a Positive, being metals they tend to lose electrons **b** Manganese **c** Only one stable oxidation state
10 a Cr^{3+}, Mn^{6+}, Fe^{2+}, Ni^{2+} **b** Mn^{2+}, Co^{2+} **c** Co^{3+}, Fe^{3+} **d** Mn^{7+} **e** V^{5+}, Cr^{6+} **f** Ti^{3+}, Fe^{6+} **g** V^{3+}, V^{4+}, Cu^{2+}
11 a Oxidised **b** Reduced **c** Oxidised **d** Oxidised **e** Reduced
12 Change in colour

19 Chromatography

1 Colour or dye **2** Concerned with legal matters/crime
3 Spot of colour is placed in the middle. Solvent is added to spot of colour. Solvent carries colours with it as it spreads through paper and makes rings
4 a Paper **b** Solvent **5** Four
6 Dyes, amino acids, pharmaceuticals, food additives
7 Distance travelled by mixture compared to that travelled by solvent

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ISBN 978 1 92561 014 7

8 Both use capillary action, separate mixtures and use solvents. Paper chromatography involves absorption and TLC involves adsorption

9 Gravity causes the solvent (eluent) to flow down the column. As it passes through the stationary phase, the components of the analyte (mixture) have different attractions for the stationary phase and hence pass through at different speeds

10 **a** true **b** false **c** false **d** false

20 Gas–liquid chromatography

1 Nitrogen, helium, argon **2** Keeps the sample in gaseous form

3 Unreactive, can withstand heat, can be bent

4 10^{-9} g **5** **a** B **b** D **c** E **d** F **e** A **f** C **g** G

6 **a** Takes analyte with it **b** Supports the stationary phase **c** Keeps contents of column in gaseous phase **d** Registers presence and amount of component **e** Adjusts flow of gas **f** Where the substance is introduced into the system **g** Makes a record of what is detected

7 To save space **8** Large collection of data that can be searched easily using specific criteria

21 Electrophoresis

1 Ion attracted to the anode (i.e. negative)

2 Ion attracted to the cathode (i.e. positive)

3 To allow movement of ions

4 Maintains pH within a narrow range

5 They are slower to move than smaller particles

6 Amount of charge and shape

7 To act as a basis to compare against others

22 The mass spectrometer

1 **a** B **b** G **c** A **d** H **e** C **f** D **g** F **h** E **3** Positive

4 So they are not slowed down or stopped by colliding with air particles **5** **a** True **b** False **c** False **d** True **e** True

23 Interpreting mass spectra

1 **a** Larger **b** Increases **2** Single positive charge

3 (206 × 24.6/100) + (207 × 22.5/100) + (208 × 52.9/100) = (50.68) + (46.58) + (110.03) = 207.3

4 12 = C, 16 = O, 28 = CO, 44 = CO_2 (Correspond to masses of these atoms/molecules)

5 1 = H, 35 = Cl-35, 36 = HCl-35, 37 = Cl-37, 38 = HCl-37

6 Cl-35, peaks for this isotope are higher than those of Cl-37

24 Nuclear chemistry

1 Atomic mass units **2** They are located in the nucleus

3 No. They are atoms of different elements because they have different numbers of protons

4 Almost all are U-238.

5

Name	Nature of radiation	Electrical charge	Speed (relative to light)	Penetrating power
Alpha	Helium nuclei	Positive	5%	Low
Beta	Electrons	Negative	100%	Low
Gamma	emr	None	100%	High

6 True **7** **a** Electrons (Beta radiation) **b** Gamma rays **c** Alpha

8 **a** $^{226}_{88}Ra \rightarrow {}^{222}_{86}Rn + {}^{4}_{2}He$ **b** $^{14}_{6}C \rightarrow {}^{0}_{-1}e + {}^{14}_{7}N$
c $^{32}_{15}P \rightarrow {}^{0}_{-1}e + {}^{32}_{16}S$

9 Energy released is the product of the mass times the square of the speed of light

10 When a neutron hits a uranium-235 atom, an atom of uranium-236 is formed which then breaks down into an atom of strontium-90, an atom of xenon-143 and three neutrons

25 Radioactivity and half-life

1 Time, radioisotope, original, disappeared, quarter, eighth

2 3.13% **4** **a** 1.7 hours **b** 15% **5** **a** False **b** True **c** False

26 Risk assessments

1 **a** Exaggerated response by the body to a foreign substance **b** Causes cancer **c** Damages materials on contact **d** Easily set on fire **e** Chemical that dulls the senses **f** Poisonous **g** Danger **h** Make as small as possible

2 **a** Handle with test tube holder, place in test tube rack. Wear lab coat and shoes with leather uppers **b** Avoid contact **c** Wear goggles **d** Keep well ventilated, use a fume cupboard **e** Keep hair tied back. Keep flammable substances well away **f** Keep well ventilated, use a fume cupboard

4 Bags pose a hazard, students could trip over them

27 Practical work

1 **a** What you are trying to find out **b** Summary of results **c** Scientific drawing of apparatus **d** Apparatus that was used **e** How you carried out the practical work **f** What happened **g** Name of the experiment

2 Hypothesis, areas that could be improved. **3** Date

4 Purpose, what you intend to do

5 Two dimensional, cross section, ruler, drawn in pencil, labelled in blue/black pen

6 Write using past tense and without personal pronouns, e.g. 'The water in the beaker was heated.'

7 Table, graph, description, diagram, photo, video recording

8 One that you use when you can't think of anything else.

28 Variables and controls

1 Size of capsule, stirred or not stirred **2** Fizzing stopped

3 Size of chips **4** Time for reaction

5 Mass of chips, volume of acid, concentration of acid, temperature

6 Smaller chips react faster than large chips

7 Calcium carbonate + hydrochloric acid → calcium chloride + carbon dioxide + water

8 $CaCO_{3(s)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + CO_{2(g)} + H_2O_{(l)}$

29 Formulae of compounds

1 **a** Hydrogen (×1), chlorine (×1) **b** Carbon (×2), hydrogen (×6) **c** Hydrogen (×2), oxygen (×2)

2 Carbon dioxide: CO_2, CO_2
Oxygen: O_2, O
Ozone: O_3, O_3
Benzene: C_6H_6, CH
Hydrazine: N_2H_4, NH_2
Ethene: C_2H_6, CH_3
Ethanoic acid: $C_2H_4O_2$, CH_2O
Ethan-1, 2 diol: $C_2H_6O_2$, CH_3O
1, 1-dibromo, 2, 2-dichloroethane: $C_2H_2Br_2Cl_2$, CHBrCl
Glucose: $C_6H_{12}O_6$, CH_2O

30 Calculating formulae

1 **a** 7.99/12 : 2.1/1 = 0.67 : 2.1 = 3.1 : 1. To the nearest whole number 3:1, hence CH_3
b 1.88/12 : 0.625/2.5/16 = 1 : 4 : 1 hence CH_4O

2 5.46/31 : 7.04/16 = 2 : 5 hence P_2O_5

3 **a** 1.29/12 : 0.108/1 = 1 : 1, hence CH, (×6) C_6H_6
b 0.84/12 : 0.14/1 : 1.12/16, hence CH_2O, (×6) $C_6H_{12}O_6$

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An imprint of Blake Education Pty Ltd
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ISBN 978-1-92561-014-7